



Data Driven Decision Making

Strategic Publishing Pathways: Enhancing the Reach of
AI-Driven Substance Use Disorder Research

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DOCUMENT CONTROL

Work carried out by:

Name	Email Address	Task description
Art Casasa	axc1068@psu.edu	Deliverable 1: Contributed to problem statement, features to take into consideration for search of best recommendation Deliverable 2: General review of content vs. Requirements for deliverable
Benn Kulcsar	exk324@psu.edu	Deliverable 1: Offered an avenue for Potential Resources, collaborated on the Problem Statement and Timeline and brainstormed Data Driven Questions, outlined the Anticipated Results Deliverable 2: Contributed to revising Deliverable 1; Reviewed and added missing insights to the database exploration Deliverable 3: Contributed to identification of biases and issues within our dataset and the description of the data within the dataset, helped editing the sections Deliverable 4: Collaborated on updating Deliverable 3; Performed cleaning of our dataset using KNIME and write-up of the steps taken Deliverable 5: Contributed to the write-up of data modeling types and plans; Described changes made to dataset for modeling Deliverable 6: Edited and expanded upon the Modeling and Biases sections Deliverable 7: Contributed to Conclusion/Next Steps, Participated in presentation.
Joseph Chieffallo	jsc5533@psu.edu	Deliverable 1: Doc preparation, brainstormed Data Driven Questions, and wrote the Feasibility Estimate Deliverable 2: Created summary statistics for class distribution evaluation Deliverable 3: Added some visualizations and identifying cognitive bias section Deliverable 4: Corrected visualizations to include Unknown venues prior to data cleaning. Added plan to answer descriptive questions Deliverable 5: Added plan to answer predictive and prescriptive questions. Added variable selection section Deliverable 6: Created Classification Tree models Deliverable 7: Participated in discussion and read a portion of the presentation

Joshua Martin	jjm6589@psu.edu	Deliverable 1: Initial ideas on processes of project, assisted in getting meeting scheduled, and participated in clarifying direction for deliverables Deliverable 2: Proofread and worked on content and clarity in document, posted comments and participated in discussion. Deliverable 3: Clarified sections of report and helped with formatting, participated in comments and concerns Deliverable 4: Continued with editing the technical writing throughout the report for clarity, accuracy, and conciseness Deliverable 5: Editorial discussions, accuracy checks, and writing on the modeling sections Deliverable 6: Checked for consistency in the document, reviewed conclusions and interpretations of model results Deliverable 7: Participated in discussion and read a portion of the presentation.
Logan Eichbaum	lke5200@psu.edu	Deliverable 1: Added to Potential Resources, made minor edits to the Problem Statement, drafted Executive Summary and Timeline. Deliverable 2: Created draft of data structure template, included comments to invite further discussion on feature space. Deliverable 3: Added visualizations, made updates to preliminary analysis Deliverable 4: Wrote about several biases introduced by cleaning the dataset, added tables to the project describing the results of cleaning the data. Deliverable 5: Added explanation and plan for using logistic regression Deliverable 6: Created model for multinomial logistic regression. Drafted analysis on the same model. Deliverable 7: Participated in presentation.
Matthew Cecchetti	mvc667@psu.edu	Deliverable 1: Drafted Problem Statement, brainstormed both the Business Question and Data-Driven Questions Deliverable 2: Made edits to the attribute values. Added the handling of null values question. Added handling empty or Null Values. Edited the feedback on the deliverable 1 problem statement. Small change in the relevant information section. Deliverable 3: Edited and ran the Python code to extract the data from the PDFs and pull them into an Excel file. Added visualizations, descriptions and edited sections. Deliverable 4: Cleaned up the Python code and reran it with all Adobe documents. Added fixes to the KNIME workflow. Added visualizations. Plans for next steps Deliverable 5: Reviewed and added edits to deliverable 5. Helped with determining which model to use. Added Transformation, Data Reduction, Data Split and Validation, Bias Control section.

Revision Sheet

		Deliverable 6: Created the Random Forest Model and section, and reviewed/edited other sections. Deliverable 7: Edits to the Conclusion and Next Steps section. Participated in presentation.
Tom Brashear	tjb7033@psu.edu	Deliverable 1: Drafted Executive Summary, collaborated on Business and Data-Driven Questions and Timeline. Deliverable 2: Contributed to editing Deliverable 1. Content additions/review/comment/edits for Deliverable 2. Deliverable 3: Reviewed code and code output. Evaluated data (articles, cited_by, cites for completeness), contributed to deliverable write up. Provided initial thoughts on data cleaning plan. Deliverable 4: Worked data completeness by ensuring all articles were downloaded and in the dataset. Provided input / review to the deliverable and worked on the biases introduced by data cleaning and expanded on the plans to clean the dataset. Deliverable 5: Added bias analysis from the data modeling preparation. General review and edit of deliverable. Deliverable 6: Revised portions of Deliverable 5 based on feedback. General contribution, review and editing of deliverable 6. Deliverable 7: Contributed to Conclusion/Next Steps. Contributed to and participated in presentation.

Revision Sheet

Release No.	Date	Revision Description
1	9/13/2025	Crafted and Submitted Deliverable 1
2	9/18/2025	Revised the Problem Statement to fix the scope of the audience and the problem we are addressing.
3	9/21/2025	Completed Analysis of Database and Revised Deliverable 1
4	10/5/2025	Scraped Data from Uploaded PDFs and Performed an Initial Analysis of the Data
5	10/18/2025	Revised the organization of Deliverable 3 and added a table
6	10/19/2025	Cleaned Updated Dataset and Addressed Related Descriptive Questions
7	11/2/2025	Revised Dataset for Modeling, Described Modeling Plans, and Re-addressed Biases within our Methods
8	11/16/2025	Revised the content of Deliverable 4, Documented Modeling Steps and Outcomes, Discussed Biases, and Revisited Predictive and Prescriptive Questions

Revision Sheet

9	12/11/2025	Added a Conclusion summarizing our Findings and Next Steps, Reviewed entire document for final edits
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PROJECT PROPOSAL

Executive Summary

The opioid epidemic continues to exert a devastating toll on urban and rural communities, with millions of individuals unable to access the treatment they urgently need. Substance Use Disorder (SUD) recovery is a long and complex process and treatment needs to “address the needs of the whole person to be successful” (*NIDA, 2020*). In response, research teams have begun to implement Artificial Intelligence (AI) algorithms to uncover connections between existing data to shed light on potential avenues for remediation or suggest improvements on current strategies. Additionally, Systematic Literature Reviews (SLR) aim to compile and summarize these discoveries. Optimizing data gathered by the factors of visibility, credibility, and impact of each potential venue will allow a data-driven approach to selecting the best place to publish. We aim for this decision to have a major impact as it directly affects the dissemination of knowledge to public health officials, AI researchers, and affected communities who rely on timely, relevant insights to guide strategic interventions.

To address this unmet need, our project proposes a data-driven framework that will recommend the most suitable journal for publishing the SLR. This framework will evaluate potential venues using a multi-criteria approach, incorporating viewership metrics (online vs. print reach), citation patterns, impact factors, accessibility (academic vs. public), and domain relevance to both AI and SUD research. By aligning the publishing strategy with stakeholder goals, we aim to enhance the reach and influence of the research, ensuring its findings can inform changes in both policy and practice. As Vaughan emphasizes in Chapter 1, “the first step in any analytical project is to define the problem in terms of a decision that needs to be made” (Vaughan 4). Our project centers around the decision: where can authors publish for maximum strategic impact.

Our technical approach will follow the CRISP-DM methodology. First, we define the business problem and how to optimize the publishing decision for strategic impact. Next, we will analyze metadata from relevant literature, including citation counts and journal classifications, to understand the landscape. The data collected will be cleaned and structured using Python and statistical tools (Scrape Scholar output), to enable robust modeling through clustering or scoring algorithms. The models will rank journals based on their alignment with stakeholder priorities. Using multiple models will allow for exploration into what techniques are more successful in forming connections relevant to our problem. Evaluation will involve validating the recommendations against real-world publishing outcomes and stakeholder feedback. Finally, the deployment phase will culminate in a comprehensive report that presents our data-driven actionable insights and visualizations. As Vaughan notes in Chapter 2, “data understanding is not just about exploring the data, it’s about connecting the data to the decision context” (Vaughan 29), which will guide our metadata analysis and feature selection.

Throughout the project, we will integrate key course concepts such as literature mining,

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feature selection, ethical considerations, and interpretability. These elements not only enhance the technical rigor of our approach but also ensure that our recommendations are transparent, defensible, and aligned with best practices in data science. Chapter 3 reinforces our belief that to follow best practices, “ethical awareness and interpretability are not optional—they are essential to building trust in analytical outcomes” (Vaughan 61). By grounding our work in the analytical principles outlined in Vaughan’s framework and the strategic needs of our stakeholders, we aim to deliver a solution that is both technically sound and socially impactful. Our recommendations for a publishing venue will aim to maximize the impact of the article on the community that is focused on mitigating and offering innovative solutions to shortcomings for SUD treatment.

The Problem: Why Publishing Strategy Matters

Authors are facing difficulties trying to find the best venue to publish their research. Authors are under significant pressure to meet their metrics (h-index, altimetric score, impact factor, and citation), which will increase their rating and enable them to secure funding (Richtig 1441). Companies have knowingly been aggressively targeting authors to publish their research in open-access journals for a fee, but these journals frequently fail to conduct proper peer review (Richtig 1441). Any journal that does not conduct appropriate peer review removes its credibility and prestige, potentially reducing the overall reach the author is trying to obtain. The focus is to provide authors who write research on Substance Use Disorder (SUD) with a recommendation on where to publish research that effectively maximizes reach and impact on their target audience. This recommendation will encompass the publishing venue that is likely to ensure visibility, relevance, prestige, and reach the appropriate target audience.

Supporting Questions

Our team explored what business question would lead to the best path to satisfy the stakeholder requirements: how can we assure increased visibility for the SLR when choosing a publishing site? This question led us to ask the following descriptive, predictive, and prescriptive questions respectively: where are relevant articles being published and what is the overall audience reach to critical research fields of these articles? Which journals are most likely to yield higher citation rates for a Systematic Literature Review (SLR) on Artificial Intelligence (AI) and Substance Use Disorder (SUD)? Given the choice, what publishing venue should authors choose to maximize the reach of their work?

If our team were to determine where relevant articles are being published and what is the overall audience reach to critical research fields of these articles, we will be able to predict which journals are most likely to yield higher citation rates for research on the intersection of AI and SUD research. Based on these results, the stakeholders would be able to maximize their publication’s visibility.

Potential Timeline

Week	Milestone	Focus Area
2	Deliverable 1	Complete the Project Proposal – Define the goal of the project, what data sources will be utilized, and what our intended results are. This piece will be the guideline for the coming weeks. Create shared document for continuing work.
3 + 4	Deliverable 2	Data Collection – Review the depth of the provided Scrape Scholar and determine what additional sources of information would be beneficial to understand the publishing venues listed. Support other teams in gathering full-text papers. Focus on identifying features that characterize venue quality (e.g., impact factor (normalized), eigenfactor score). Continue progress in the shared document for the final report; upload all drafted sections.
5 + 6	Deliverable 3	Data Understanding – Format initial data scrape and analyze citation and viewership trends and comprehend what additional features (if any) should be added for analysis throughout both weeks. Identify barriers to analyzing citation/viewership trends. Complete the introduction and begin working on the related work portions of the final report.
7 + 8	Deliverable 4	Data Cleaning – Significant effort will be put into formatting, cleaning, and normalizing data to ensure our intended training parameters are not misinterpreted by our algorithm and begin exploring what techniques we can employ with different algorithms during both weeks. Complete the related work and amend new info to the approach and experimental sections of the final report.
9 + 10	Deliverable 5	Data Preparation – Ensure dataset is cleaned and prepare features for modeling on week 9. Finalize modeling techniques and parameters in week 10. Complete the Study Design and Methodology portion of the final report. Continue to amend new info to the approach and experimental sections of the final report.
11 + 12	Deliverable 6	Model Refinement– Create and utilize ranking models and tweak parameters to get the best results without creating skewed data in week 11. Evaluate the models and determine which technique is the best fit to answer our business question in week 12. Complete the Data Analysis and Results portion of the final report. Complete the approach and experimental sections of the final report.

13 + 14	Deliverable 7	Final Draft – Reassess initial goals and craft the conclusion to tie the whole project together. Ensure all business and analytical questions are answered. Finish the Final report during the last two weeks.
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Potential Resources

Our model and analysis will be based on a dataset of selected relevant papers, all pertaining to the promise of AI modeling methodologies and relevancy to understanding the factors of SUD. If needed, we will further explore additional analytics of each publishing venue to better understand if there are other factors which make a venue more advantageous for publishing.

Feasibility Estimate

The risk assessment matrix below was developed to help the team better understand the challenges that are most likely to be faced going forward. For each item, impact and likelihood were graded on a scale of 1-5: these two scores were multiplied to assign a comparable impact score for each risk. Risks with both low severity and likelihood (with scores less than 3) were omitted.

Item	Risk	Severity	Likelihood	Impact
1	Insufficient Number of Usable Observations in Initial Dataset	5	3	15
2	Difficulty In Creating Successful Models	3	2	6
3	Lack of Coordination among Team Members	4	1	4

The team's mitigation strategy for item 1 is to create multiple iterations of the training dataset, each time experimenting with different weighting of the variables identified as important with the goal of creating actionable connections between publishers and visibility to the scientific community. If the dataset is particularly large, formatting and cleaning might prove to be a lengthy process. In this case it would either need to shrink or risk variables being weighted inequality in the model, skewing intended results. For item 2, the team plans to create and optimize multiple types of machine learning models to explore how different algorithms predict the value provided by publishing in a particular venue. For item 3, the observed coordination between team members occurring in the Microsoft Teams group has proved to be sufficient so far, and ensuring continual open lines of communication between team members is the team's best way to mitigate this risk.

Anticipated Results

At the end of the term, we expect our project to: 1. Transform the Scrape Scholar database of articles into effective training data for multiple machine learning models, allowing the user to determine what publishing venue would offer the highest visibility for Substance Use Disorder research, 2. Create multiple machine learning models on our database and generate a report on

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each, including successes and shortcomings. 3. A detailed report of every step and justifications for our decisions as well as an extensive recommendation section based on all our analysis.

DATASET DESCRIPTION AND EXPLORATION

Title: Article Visibility Assessment Database

Sources:

(a) Creator: Group 2

(b) Date: September 20, 2025

Past Usage:

This database was created as a subset of a larger Scrape Scholar spreadsheet created by students and faculty at Penn State.

Relevant Information:

The Article Visibility Evaluation Database was created as a subset from the larger Scrape Scholar database. It contains features that are relevant in determining how much of an outreach an article published in a scientific venue. These features include:

Date	date article was published
Cited	number of times the article was cited
Methodology	
Clarity	
Transparency	
Relevance	

The date of the article will be weighted in analysis to determine how relevant the article is relative to the present; articles that have high visibility but are outdated should not contribute as much to the visibility as current articles. The number of citations conveys how many times the article was used, therefore how much of an outreach it has. Methodology, Clarity, Transparency, and Relevance are all binary variables that indicate how closely related the article is to SUD factors.

A first glance at the data reveals that the date format is not standardized throughout the spreadsheet. This will be adjusted to include only the year of the article before conducting analysis.

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There are 28 rows that have empty values for Methodology. 29 rows have empty values for Clarity, Transparency, and Relevance.

Number of Instances: 428

There are 428 individual rows in the dataset – each with a unique article ID and identifying information pertaining to that article.

Number of Attributes: 7 of interest

Attribute Values:

Date	1/12/2025, 31/12/2025, 2024, more	Continuous
Cited	1, 2, 37, more	Numeric
Methodology	0, 1	Binary
Clarity	0, 1	Binary
Transparency	0, 1	Binary
Relevance	0, 0.5, 1	Ordinal
Source	ScienceDirect, Scopus, BMC, more	Categorical

Missing Attribute Values:

Methodology, Clarity, Transparency, and Relevance are missing values for the same article's rows 145 and 170 through 195. Article 144 is only missing Clarity, Transparency, and Relevance. Row 295 has no value in any of its columns.

Cited: 1 missing row

Methodology: 28 missing rows

Clarity: 29 missing rows

Transparency: 29 missing rows

Relevance: 29 missing rows

Source: 1 missing row

Date: 1 missing row

Class Distribution:

The following table was generated to better understand the distributions in the Scrape Scholar dataset. Fields of interest are highlighted below:

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Table 1: Scrape Scholar Interesting Findings (Abbreviated)

Source	Number of Publications per Source	Percent of Total Publications per Source	Total Citations per Source	Average Citations per Source	Percent of Total Citations per Source
ScienceDirect*	171	40.14	2703	15.81	25.03
Scopus*	159	37.32	1093	6.87	10.12
BMC	2	0.47	1470	735.00	13.61

**Scopus is a scientific abstract and citation database and ScienceDirect is a searchable web-based bibliographic database, making neither a publisher, just where the article was retrieved from by the Scrape Scholar tool. These publications gathered from these sources will need to be further investigated so that the true publishing entities can be evaluated.*

Ideally, the different classes would have roughly equal representation in the dataset to allow for a fair comparison between classes. Initial evaluation of the Scrape Scholar data revealed several trends in distribution that are far from ideal. The original dataset contains observations from 52 unique sources: of these, 2 sources published over 77% of the total number of publications listed in this dataset.

While it is reasonable to expect some varying degree of popularity between each publication, the citation distribution is extreme. In total, the articles in the dataset have been cited 10,799 times. The top 15 most cited publications contribute to more than half of the total number of citations in the dataset. Other sources in the dataset include a very low number of publications with a very high average number of citations per publication, indicating outliers are present.

Handling Empty or Null Values:

Empty or null values will be removed from our analysis. The missing data occurs because the values for those articles simply do not exist. This is why we are characterizing the missing data as structurally missing data. The analysis can only be run on values pulled from each article. Since these articles do not have a value, they need to be excluded. The downfall of removing the lines is that the sample size decreases, but this eliminates the bias of filling in the empty values. The goal of this project is to identify the best venue to publish journals based on the collected data. Replacing any missing values with zero or an average value would modify our findings and conclusions.

PRELIMINARY ANALYSIS

Building the In-house Dataset

Before extracting any data, our team had to determine what features might be useful in our analysis. Our conclusion after delving into the Scrape Scholar output was that we should consider more features than just the publishing venues in the scope of the analysis. Features such as document type and reference count have the potential to uncover more insights and provide a more informative and nuanced conclusion than analysis along a single dimension.

The Scrape Scholar dataset was missing a field for venue, which is the most important field for the analysis. Keeping in mind the primary goal of the analysis is to recommend an optimal venue; the venue name needed to be imported into the dataset while removing fields that are no longer required, allowing the dataset to contain only the necessary fields. To build the dataset, Python was used to extract information from the provided article PDFs. The process extracted reference_count, pages, and digital object identifier (DOI). While pulling the data, the process required a method to retrieve the venue information accurately, as this is vital for the analysis. Therefore, the process utilized metadata sources using the DOI. The code first checks metadata sources, which are in the order Crossref, DataCite, OpenAlex, and Elsevier, using the DOI from the PDF and referencing the identification number from their respective databases. If there is no match, then the process runs a title search and other identifiers from the PDF file. If there is still no match, the venue value is marked as “Unknown”. Once the process was able to pull the venue information, the next step was to combine the new PDF information with the original Scrape Scholar data. For the data table, the process does not need all the data provided by Scrape Scholar, so some columns were removed. The following columns were kept for the upcoming analysis:

Table 2: In-House Dataset Features

Column Title	Source	Data Type	Description
id	Scrape Scholar	Numeric	Unique numeric identifier for each observation
title	Scrape Scholar	Categorical	The title of the document, used for text-based features such as topic modeling and readability
date	Scrape Scholar	Numeric	Date the article was published, with some columns containing timestamps
year	Scrape Scholar (modified)	Numeric	Year value stripped from Scrape Scholar date column
citedby	Scrape Scholar	Numeric	Number of citations received; a proxy for article impact
Methodology	Scrape Scholar	Binary	Rubric-based evaluation dimensions, indicating perceived document quality
Clarity	Scrape Scholar	Binary	
Transparency	Scrape Scholar	Binary	
Relevance	Scrape Scholar	Binary	

Score	Scrape Scholar	Numeric	Composite rubric score of Methodology, Clarity, Transparency, and Relevance
document_type	Scrape Scholar	Categorical	Type of published document
venue	metadata sources	Categorical	Document's publishing venue
reference_count	PDF	Numeric	Number of references within the document
pages	PDF	Numeric	Number of pages of the document

Not all PDF files returned with a venue name after being processed through the Python code, leaving an “Unknown” value in place. Methodology, Clarity, Transparency, and Relevance are all categories determined prior to this analysis and assess an article’s integrity and alignment with AI and Substance Use Disorder research. Values in these columns are binary, with 1 indicating an article fulfilled the requirement and 0 indicating it did not. The relevance of these metrics will be determined in a later analysis.

Despite only needing to focus on the venue, there are a few prominent issues with our dataset. Besides not having all the initial data, the data will need to be cleaned after ingesting it to account for characters that are not able to be parsed or nonsense. As it stands, our team is unsure if the data provided will be enough to form definitive and insightful analyses. We believe supplemental information might be necessary, such as measures like the number of downloads or the number of views on each article. Although our scope does not venture outside of the Scrape Scholar results, we are forced to reckon with what our current dataset can uncover about the reach of the article.

The dataset has 240 distinct values inside the venue column. Out of those 240 distinct venue names, there is one unknown value, which has 20 rows in total. As shown in Table 3 below, “Unknown”, “Drug and Alcohol Dependence”, and “Addiction Science & Clinical Practice” lead the venue column as the most shown in the dataset.

Table 3: Distinct Venues with Count of Rows

ID	Venue	Count
1	Unknown	20
2	Drug and Alcohol Dependence	16
3	Addiction Science & Clinical Practice	9
4	Scientific Reports	8
5	Journal of Surgical Research	7
6	BMC Public Health	6
7	Contemporary Clinical Trials	6
8	PLOS ONE	6
9	Health & Justice	5
10	Journal of Medical Internet Research	5
11	Journal of the American Medical Informatics Association	5

12	Addictive Behaviors	4
13	Heliyon	4
14	JAMA Network Open	4
15	Philosophy & Technology	4
16	Psychiatry Research	4
17	npj Digital Medicine	4
18	American Journal of Epidemiology	3
19	Anesthesia & Analgesia	3
20	Annals of Emergency Medicine	3
21	Applied Sciences	3
22	Arthritis Care & Research	3
23	Arthroscopy: The Journal of Arthroscopic & Related Surgery	3
24	BMC Anesthesiology	3
25	BMC Medical Education	3
26	BMC Psychiatry	3
27	BMJ Open	3
28	BioData Mining	3
29	Communications Biology	3
30	Harm Reduction Journal	3
31	Health Informatics Journal	3
32	Health Policy and Technology	3
33	IEEE Journal of Biomedical and Health Informatics	3
34	International Journal of Drug Policy	3
35	Knee Surgery, Sports Traumatology, Arthroscopy	3
36	Neuropharmacology	3
37	Neuroscience & Biobehavioral Reviews	3
38	Preventive Medicine	3
39	The American Journal of Emergency Medicine	3
40	AORN Journal	2
41	Addiction	2
42	Alzheimer's Research & Therapy	2
43	American Heart Journal	2
44	American Journal of Preventive Medicine	2
45	Anesthesiology Clinics	2
46	Applied Materials Today	2
47	BMC Geriatrics	2
48	BMC Medical Informatics and Decision Making	2
49	Biological Psychiatry	2
50	CHI Conference on Human Factors in Computing Systems	2
51	Child and Adolescent Psychiatry and Mental Health	2
52	Choice Reviews Online	2
53	Computer Methods and Programs in Biomedicine	2
54	Connected e-Health: Integrated IoT and Cloud Computing	2

55	Digital Healthcare in Asia and Gulf Region for Healthy Aging and More Inclusive Societies	2
56	Discover Sustainability	2
57	Emerging Trends in Drugs, Addictions, and Health	2
58	Experimental Neurology	2
59	Forensic Science International	2
60	Frontiers in Artificial Intelligence	2
61	Information Sciences	2
62	Intelligent Medicine	2
63	International Journal of Medical Informatics	2
64	Journal of Biomedical Informatics	2
65	Journal of General Internal Medicine	2
66	Journal of Oral Biology and Craniofacial Research	2
67	Journal of Shoulder and Elbow Surgery	2
68	Journal of Substance Use and Addiction Treatment	2
69	Machine Learning with Applications	2
70	Mayo Clinic Proceedings: Digital Health	2
71	North American Spine Society Journal (NASSJ)	2
72	PEC Innovation	2
73	Pharmacological Research	2
74	Placenta	2
75	Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems	2
76	Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics	2
77	Proceedings of the ACM on Human-Computer Interaction	2
78	Schizophrenia	2
79	Smart Health	2
80	Surgery Open Science	2
81	Surgery for Obesity and Related Diseases	2
82	The Journal of Pediatrics	2
83	The Lancet Regional Health - Americas	2
84	2024 IEEE 6th Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC)	1
85	24th Pacific Asia Conference on Information Systems (PACIS 2020)	1
86	Academic Forensic Pathology	1
87	Acta Pharmaceutica Sinica B	1
88	Advances in Hospitality, Tourism, and the Services Industry	1
89	Advances in Medical Technologies and Clinical Practice	1
90	Ageing Research Reviews	1
91	American Journal of Obstetrics and Gynecology	1
92	Archives of Public Health	1
93	Arthroplasty Today	1
94	Artificial Intelligence Chemistry	1
95	Artificial Intelligence Review	1

96	BMC Health Services Research	1
97	BMC Infectious Diseases	1
98	BMC Medicine	1
99	BMC Pediatrics	1
100	BMC Psychology	1
101	Best Practice & Research Clinical Anaesthesiology	1
102	Biochemical and Biophysical Research Communications	1
103	Biomedical Signal Processing and Control	1
104	Biomedicine & Pharmacotherapy	1
105	Biometrics	1
106	Bioresources and Bioprocessing	1
107	Blood	1
108	Borderline Personality Disorder and Emotion Dysregulation	1
109	British Journal of Anaesthesia	1
110	Cell Communication and Signaling	1
111	Chemical Reviews	1
112	Chemico-Biological Interactions	1
113	Clinical Lymphoma Myeloma and Leukemia	1
114	Clinical Spine Surgery	1
115	Computer Law & Security Review	1
116	Computer Science Review	1
117	Computers and Electrical Engineering	1
118	Computers in Biology and Medicine	1
119	Computers in Industry	1
120	Current Behavioral Neuroscience Reports	1
121	Current Clinical Microbiology Reports	1
122	Current HIV/AIDS Reports	1
123	Current Opinion in Biomedical Engineering	1
124	Current Pain and Headache Reports	1
125	Current Pharmacology Reports	1
126	Current Treatment Options in Psychiatry	1
127	Current Urology Reports	1
128	Decision Support Systems	1
129	Disease-a-Month	1
130	Emergency Medicine Clinics of North America	1
131	Encyclopedia of Forensic and Legal Medicine	1
132	Frontiers in Psychiatry	1
133	Frontiers in Public Health	1
134	Fundamental Research	1
135	Gastroenterology	1
136	Health & Justice	1
137	Health Care Management Science	1
138	Healthcare Analytics	1

139	IEEE Access	1
140	IEEE Internet of Things Journal	1
141	IEEE Transactions on Big Data	1
142	IFAC-PapersOnLine	1
143	INQUIRY: The Journal of Health Care Organization, Provision, and Financing	1
144	Implementation Science	1
145	Implementation Science Communications	1
146	Interdisciplinary Journal of Information, Knowledge, and Management	1
147	International Journal of Environmental Research and Public Health	1
148	International Journal of Information Management Data Insights	1
149	International Journal of Mental Health Systems	1
150	International Journal of Neural Systems	1
151	International Journal of Obstetric Anesthesia	1
152	International Review of Economics & Finance	1
153	Irish Journal of Medical Science (1971 -)	1
154	JAAOS: Global Research and Reviews	1
155	JMIR Formative Research	1
156	JMIR Medical Informatics	1
157	JMIR Perioperative Medicine	1
158	JMIR mHealth and uHealth	1
159	Journal of Addictive Diseases	1
160	Journal of Biomedical Physics and Engineering	1
161	Journal of Black Studies	1
162	Journal of Chemical Information and Modeling	1
163	Journal of Chromatography B	1
164	Journal of Controlled Release	1
165	Journal of Eating Disorders	1
166	Journal of Image and Graphics	1
167	Journal of Medical Systems	1
168	Journal of NeuroInterventional Surgery	1
169	Journal of Neuroimmune Pharmacology	1
170	Journal of Neurosurgery: Spine	1
171	Journal of Orthopaedic Reports	1
172	Journal of Orthopaedic Surgery and Research	1
173	Journal of Orthopaedics	1
174	Journal of Pain and Symptom Management	1
175	Journal of Pediatric Orthopaedics	1
176	Journal of Pharmaceutical Health Care and Sciences	1
177	Journal of Psychiatry & Neuroscience	1
178	Journal of Public Health Management & Practice	1
179	Journal of Substance Abuse Treatment	1
180	Journal of Translational Medicine	1
181	Journal of Veterinary Behavior	1

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182	Journal of the American Academy of Orthopaedic Surgeons	1
183	Journal of the American Medical Informatics Association (JAMIA)	1
184	Journal of the American Pharmacists Association	1
185	Knowledge and Information Systems	1
186	Korean Journal of Applied Statistics	1
187	Machine Learning-Based Modelling in Atomic Layer Deposition Processes	1
188	Medical Care Research and Review	1
189	Medicine in Novel Technology and Devices	1
190	Molecular Biology Reports	1
191	Molecular Genetics and Metabolism	1
192	Molecular Psychiatry	1
193	Molecular Therapy Nucleic Acids	1
194	Nano-Micro Letters	1
195	NeuroImage: Clinical	1
196	Neurobiology of Stress	1
197	Neuromodulation: Technology at the Neural Interface	1
198	New England Journal of Medicine	1
199	Nova Science Publishers	1
200	PLOS Computational Biology	1
201	PLOS Digital Health	1
202	Pain	1
203	Patterns	1
204	Pharmacology Research & Perspectives	1
205	Proceedings of the 30th ACM International Conference on Information & Knowledge Management	1
206	Progress in Cardiovascular Diseases	1
207	Progress in Molecular Biology and Translational Science	1
208	Psychiatric Services	1
209	Psychoanalysis and Popular Culture (Routledge)	1
210	Regional Anesthesia & Pain Medicine	1
211	Regulatory Toxicology and Pharmacology	1
212	ResearchGate	1
213	Safety Science	1
214	Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine	1
215	Scientific African	1
216	Seminars in Pediatric Surgery	1
217	Social Science & Medicine	1
218	Sports Medicine - Open	1
219	Substance Abuse	1
220	Surgery	1
221	Surgical Clinics of North America	1
222	Synthesis Lectures on Mobile & Pervasive Computing	1
223	Techniques in Vascular and Interventional Radiology	1

224	The American Journal of Drug and Alcohol Abuse	1
225	The American Journal of Managed Care	1
226	The American Journal of Surgery	1
227	The Egyptian Journal of Neurology, Psychiatry and Neurosurgery	1
228	The Egyptian Rheumatologist	1
229	The Journal of Arthroplasty	1
230	The Journal of Pharmacology and Experimental Therapeutics	1
231	The Journal of The Institute of Image Information and Television Engineers	1
232	The Lancet Digital Health	1
233	The Spine Journal	1
234	Therapies	1
235	Thirty-Third International Joint Conference on Artificial Intelligence {IJCAI-24}	1
236	Toxicology Reports	1
237	TrAC Trends in Analytical Chemistry	1
238	University Economic Bulletin	1
239	Vaccine	1
240	World Neurosurgery	1

Visualizing the Dataset

After gathering the needed information, the most popular venues were found by aggregating the dataset by venue. While Table 3 displays all 240 venue values, Figure 1 below shows the top 15 citations, highlighting the average citation count.

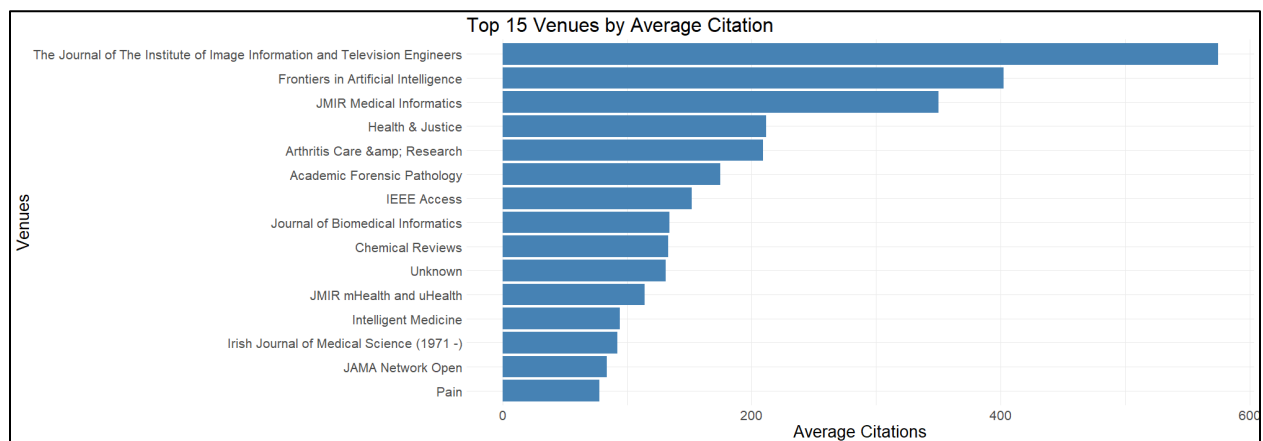


Figure 1: Average number of citations per Venue (Top 15).

Table 4 shows the number of documents published by the 15 venues with the greatest average citations as listed in Figure 1.

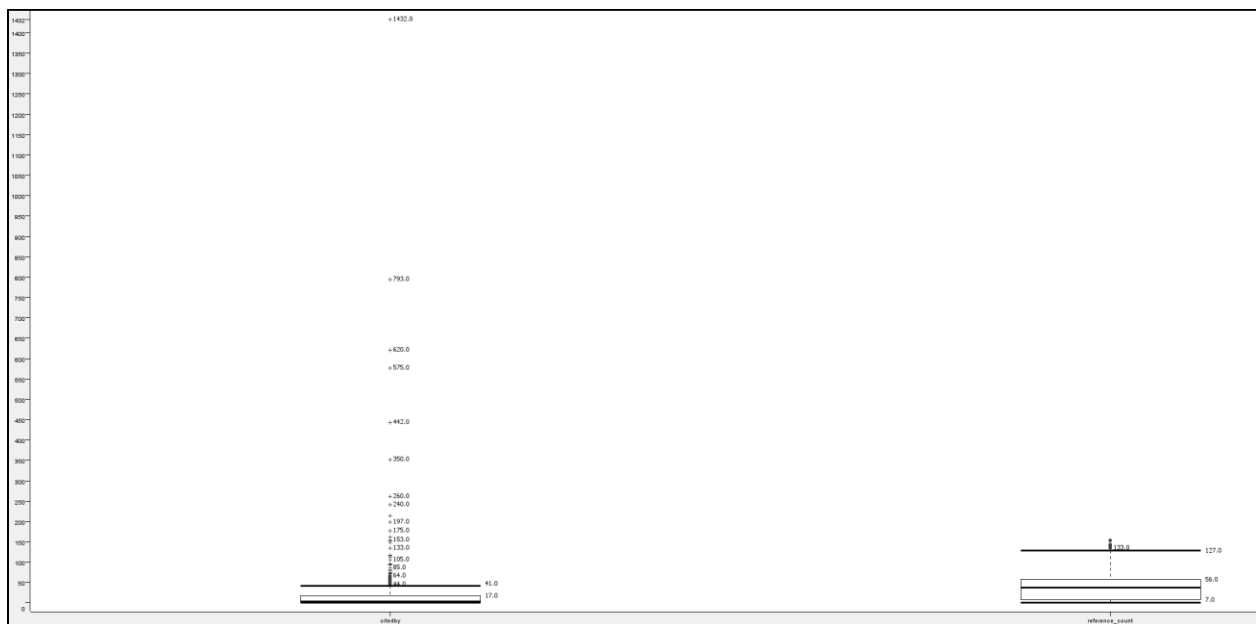
Table 4: Number of Scrape Scholar Document Publications

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Venue	Number of Publications
The Journal of The Institute of Image Information and Television Engineers	1
Frontiers in Artificial Intelligence	2
JMIR Medical Informatics	1
Health and Justice	1
Arthritis Care & Research	3
Academic Forensic Pathology	1
IEEE Access	1
Journal of Biomedical Informatics	2
Chemical Reviews	1
Unknown	18
JMIR mHealth and uHealth	1
Intelligent Medicine	2
Psychiatric Services Irish Journal of Medical Science (1971 -)	1
JAMA Network Open	4
Pain	1

As seen in Table 4, a majority of the top venues only have fewer than 3 published documents. A minimum number of Scrape Scholar publications will need to be added to ensure validity of each venue's publication sample size.

To get a better picture of the data and identify outliers in the more variable data categories, we used box plots on the fields citedby and reference_count, as shown in Figure 2 below. This provides a more intuitive understanding of the distribution of data in the two mentioned fields to highlight any extreme values.



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Figure 2: Boxplot of citedby and reference_count

Looking at Figure 2 above, we can see that the median citedby is two articles, while the upper quartile is 41. We also observe that the data contains numerous outliers. The highest outlier observed is 1,432 articles cited. The lowest value in the measure citedby is 0. When analyzing the field reference_count, the visualization shows 36 as the median count of references, and 127 is the upper quartile. The graph also displays 51 as the highest outlier in the field. The lowest value is 0.

A preliminary overview of the data reveals that most of the documents have been published in the past few years. This unfortunate, though not unexpected insight indicates that there is a temporal bias within our dataset. New articles have not had time to accumulate as many citations as they may have if they had been circulating for a longer time. While this is always a consideration since even very old papers can occasionally gain new citations in the modern day, we assume that the critical mass of citations that a paper will receive occurs within the first several years of its publication. As seen in Figure 3a below, a majority of the articles in the dataset are from 2025 at 156, with 2024 and 2023 having 73 articles and 57 articles respectively. This is so recent as to have barely had time to accumulate the bulk of citations a paper is likely to achieve throughout its lifetime.

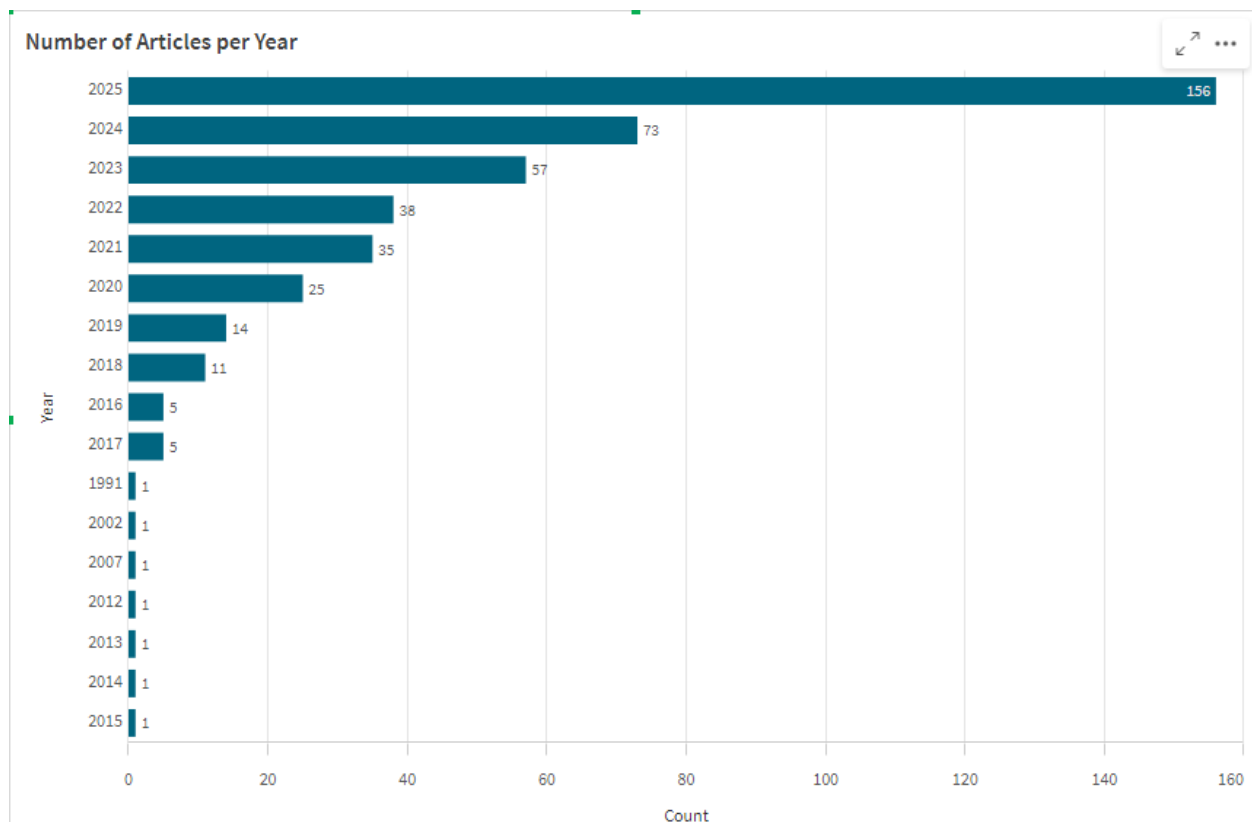


Figure 3a: Number of Articles by Year

To confirm the assumption that a disproportional number of recent articles might have introduced both cognitive and temporal biases, skewing the citation data, we will further analyze the average citations per year and use this information to determine whether to standardize the

Data Driven Decision Making

number of citations to reduce influence held by each year or remove larger outliers. Figure 3b displays the average article citations by year.

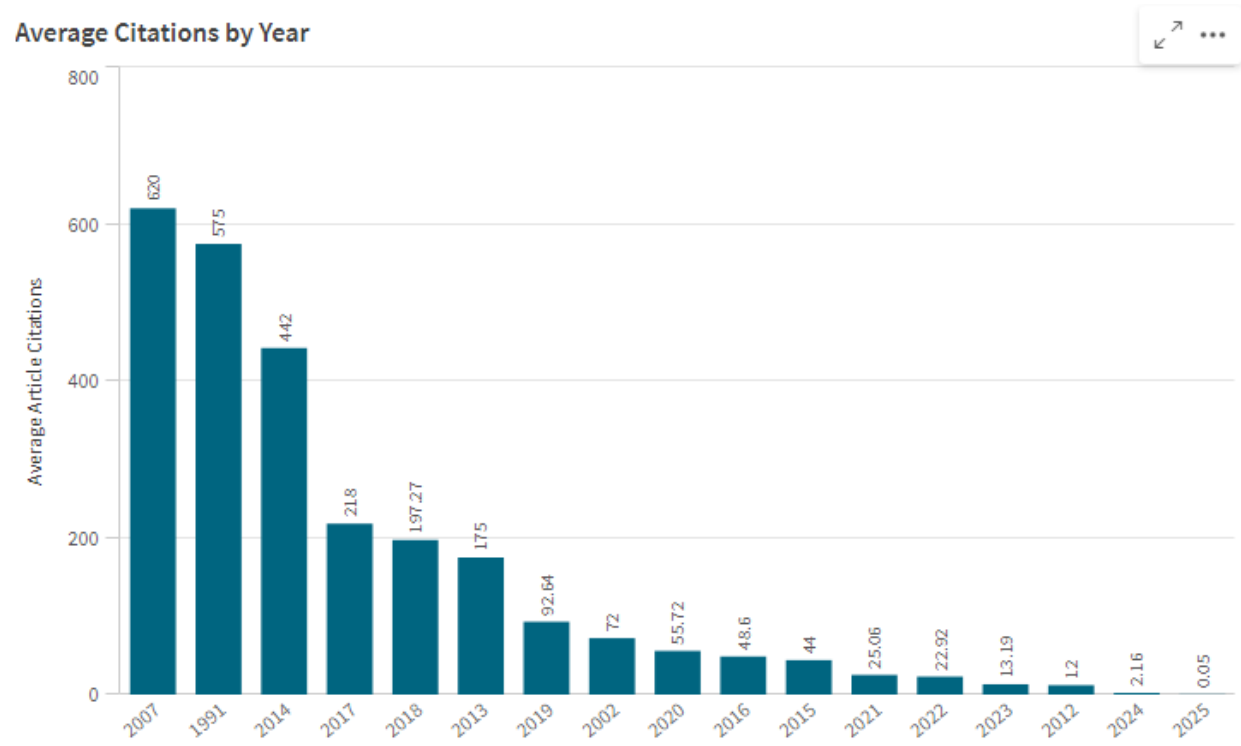


Figure 3b: Average Article Citations by Year

As shown in Figure 3a, there is only one article in the dataset from 2007, one from 1991, and one from 2014; yet, they have the highest average citation counts out of all the 428 articles as seen in Figure 3b. An additional analysis will need to be performed before ultimately deciding whether it is preferable to remove the outliers entirely or standardize them with the rest of the data to ensure no important metrics are lost.

Table 5: Percentages of Null Values Inside of Each Column

id	0.000000
title	0.467290
date	0.467290
year	0.467290
citedby	0.467290
Methodology	6.775701
Clarity	7.009346
Transparency	7.009346
Relevance	7.009346
Score	70.794393
document_type	0.467290
venue	0.000000
reference_count	0.700935
pages	0.700935

The results shown in Table 5 indicate that the score column contains a very large number of Null values at over 70 percent, especially when compared to the other counts. A simple recalculation of the four columns (Methodology, Clarity, Transparency, and Relevance) would dramatically reduce the nulls in the field. Table 5 also shows that the features derived from the PDFs, specifically `reference_count` and `pages`, have a similar percentage of missing values at 0.7%. Table 5 also shows that the venue has no null values by design as missing values are indicated as “Unknown” rather than null. This makes sense since not all the articles are yet available to scrape with the tool.

Through our extraction of data from the Scrape Scholar dataset, we encountered a large number of null or missing values in features such as date, cited by, venues, and reference count. These are values that can be input manually later to ensure data integrity. Although this results in loss of data, it is not a loss that will impact any findings when finding venues for publication.

Identifying Cognitive Biases

Cognitive biases, whether known or subtle, can affect the analyses of data. According to Dobelli’s novel *The art of thinking clearly*, responsible researchers should also reflect on their own biases and how it might affect the data they work on. Selection and publication bias skew the dataset toward high-impact, positive-result studies, while cognitive biases may influence subjective scoring fields like Clarity and Relevance. Temporal bias is also present, as newer articles haven’t had time to accumulate citations, while older studies have had more opportunity to conduct further research and therefore more citations. To address these gaps, enriching the dataset with features like abstract text, Altmetric scores, author metadata, and funding disclosures would enable more nuanced analysis.

One major cognitive bias that will be addressed is confirmation bias. It is crucial not to assume that answering the business question (how can we assure increased visibility for the SLR when choosing a publishing site?) posed for this study will yield statistically significant results. To overcome this bias, the team must explore the data from a state of curiosity rather than a state of certainty and not misinterpret contradicting data as noise or outliers.

Another bias-related hurdle that is intrinsic to this data is recency bias. As seen in figure 3a, a vast majority of the Scrape Scholar articles have been published within the last few years. While the recent boom in popularity regarding the topic of artificial intelligence may explain why the Scrape Scholar data is skewed to include more recently published documents, the founding principles regarding many methodologies were developed and implemented in the 20th century (Anastasiia, 2025). While this may make relevant documents rarer based on the number of years that have passed since the publishing date, one could make the argument that data based on more recently published documents would be more relevant to answering the business question since the publishing environment for future publications will more accurately resemble the publishing environment of recent publications as opposed to older ones.

By proactively identifying and addressing these biases, our framework aims to produce recommendations that are statistically robust, ethically sound, and strategically relevant. This aligns with Vaughan's principle that "ethical awareness and interpretability are not optional, they are essential to building trust in analytical outcomes" (Vaughan, Ch. 3). Periodically, we aim to revisit these identified biases and reexamine if we managed to avoid including the biases in our work.

Plans for Cleaning the Dataset

To perform data cleaning, our team is proposing a structured plan designed to address the specific data quality issues identified during our preliminary analysis. This plan aims to preserve data integrity while maximizing the usable sample size and ensuring that our recommendations are based on reliable, consistent data. The cleaning strategy is organized around five key steps established during our initial data assessment and structure preparation:

1. **Handling Missing and Null Values** - Many of the original data points lack values, particularly in the score column, which will need to be recalculated. Additionally, null or empty values across other fields must be addressed to ensure completeness. This includes standardizing string fields where missing formatting or extraneous characters hinder downstream analysis.
2. **Data Standardization and Normalization** - Document titles and venue names often contain inconsistent formatting or unneeded characters. Standardizing these fields is essential to ensure that identical venues or documents are not misclassified as distinct entries. Venue names will be normalized to a consistent naming convention to support accurate aggregation.
3. **Duplicate Detection and Removal** - Once string values are standardized, we will identify and remove duplicate entries. This is especially important for venues that appear across multiple years or document types.
4. **Removing Columns and Aggregation of Data** - The first structural change will involve removing columns deemed unnecessary for regression analysis. For example, specific dates will be dropped in favor of year-level granularity. Aggregating the dataset by venue, year, and document type will also be applied to consolidate data at the appropriate level, reducing noise and improving model readiness.
5. **Outlier Assessment and Treatment** - Our visualizations revealed extreme outliers in citation and reference counts. Rather than removing these entries from an already limited dataset, we plan to categorize these attributes to minimize their impact on regression modeling. This approach balances statistical rigor with the need to retain valuable data points.

In addition to these core steps, we will also address citation correction and categorical standardization to further enhance dataset quality. Any unforeseen changes will be handled with the guiding principle of retaining as much information as possible while ensuring analytical reliability.

PREPARING THE DATA

Cleaning Procedure

Our selected cleaning application was KNIME, as it offers an easier step-by-step look at the data after each dataset transformation, and many of our group members have had previous experiences using KNIME. Its low-code/no-code process also allowed us to make iterations and changes quicker than the PANDAS package. The nodes we utilized to create our cleaned dataset will be discussed below and can be seen in Table 5.

The first node we employed loaded in our created dataset, setting up the dataset integrity before transformation. Using the *Column Filter* node, we removed the “date” and “Score” columns as discussed in the previous deliverable. Next, to be able to add the values in the “Methodology”, “Clarity”, “Transparency”, and “Relevance” columns, we needed to make sure all four attributes were integers. Since the “Relevance” attribute was scraped as a double type, we needed a *Double to Integer* node to convert it. Now we can use the *Column Expressions* node to add the four attributes and recreate the “Score” attribute.

To tackle standardizing the “title” and “venue” columns, a few *String Manipulation* nodes had to be implemented to remove any character that is not a letter, number, or space, convert the case of each word to capital, and remove any spaces before the character in the string. Afterwards, we rearranged some of the columns for ease of checking the dataset after running the workflow to see if each node worked as intended. Before continuing, we created a fork(?) in the process and used a *GroupBy* and *Duplicate Row Filter* node to check if all our venues had the same naming convention. Next, we noticed the ampersand character did not translate correctly into our created dataset and had to use another *String Manipulation* node to change the string value ‘&’ into the character ‘&’. This was easier than applying an overall two step string manipulation to remove both the word ‘amp’ (which could have resulted in the loss of parts of words) and the character ‘;’ (which could have been an important part of a document title).

After tackling the string values, we removed the last row of the dataset with the *Row Filter* node, as the algorithm that Scrape Scholar runs included an empty “Snowballing” row. To handle missing values in any integer column, the *Missing Value* node was employed to input a zero to act as missing knowledge. Thinking forward, our group wanted to remove variables that would impact the accuracy of our regression results. As our focus is on the venue in which to publish, we did not see the benefit of keeping any documents where the venue was unknown. These rows were removed by the *Rule-based Row Filter* node.

To remove document duplicates, we had to rank all the instances as either: 1 for unique or selected values, 2 for duplicates, and 3 etc. for additional duplicates of documents. Using the *Rank* node, we created a new attribute “rank” and assigned the 1s to duplicates that had the highest “citedby” and “year” value to have the most up-to-date version. With that, we were able

to easily remove any row with a rank higher than one (using the *Rule-based Row Filter* node) and subsequently remove the “rank” column using the *Column Filter* node. Now our dataset is fully cleaned.

To prepare the dataset for running regression models on, our team decided to categorize both the number of citations and number of references a document has to minimize the impact outliers in these attributes. As identified in our previous visualizations, there are many extreme outliers, but we wanted to avoid removing any entries from an already small dataset. In order to create our categorical values, we first had to group the dataset by the year, venue name, and document type using the *Groupby* node. Then, to assign values into categories, we used two *Rule Engine* nodes and specified what range we wanted for each of our numerical values. A new attribute was created for each transformation; “citedby_cat” and “reference_cat” respectively. We chose to keep only four values for each new categorical attribute based on the distribution of values in the original column (see Table 6). All that was left was to export the dataset as a new table using the *Excel Writer* node.

Table 5: KNIME Data Cleaning Structure

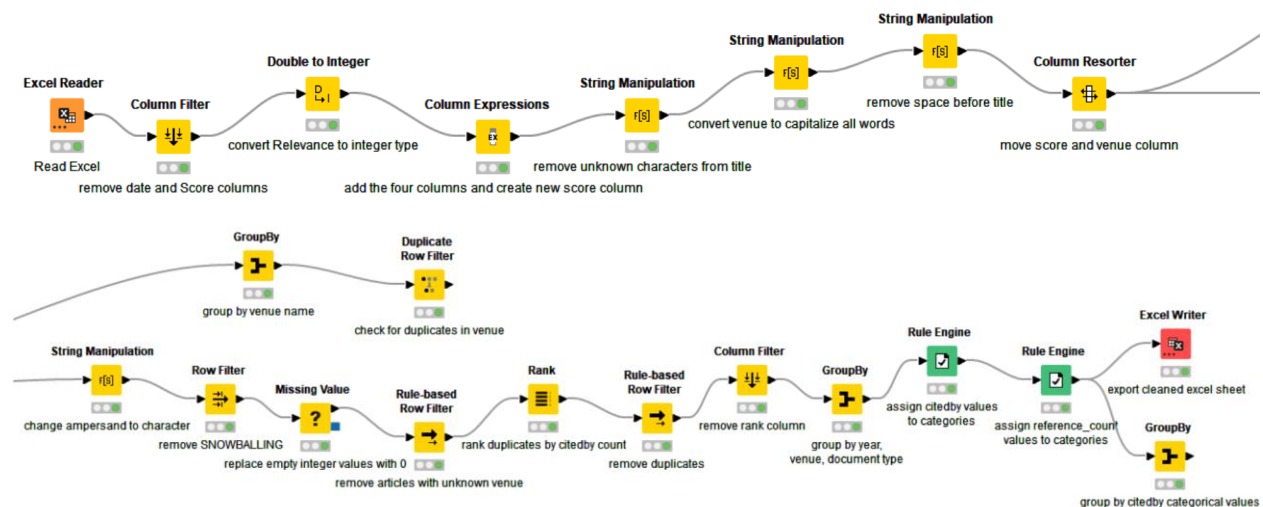


Table 6: Created Categorical Columns’ Rules

Cited By Categorical Rules		Referenced By Categorical Rules	
Range of Values	Corresponding Category	Range of Values	Corresponding Category
$0 \leq x < 5$	“0-4”	$0 \leq x < 15$	“0-14”
$5 \leq x < 10$	“5-9”	$15 \leq x < 30$	“15-29”
$10 \leq x < 15$	“10-14”	$30 \leq x < 45$	“30-44”
$15 \leq x$	“ ≥ 15 ”	$45 \leq x$	“ ≥ 45 ”

Table 7: Number of null values before and after cleaning

	Before_Column	Null_Count		After_Column	Null_Count
1	id	0	1	year	0
2	title	2	2	venue	0
3	date	2	3	document_type	0
4	year	2	4	citedby	0
5	citedby	2	5	Methodology	0
6	Methodology	29	6	Clarity	0
7	Clarity	30	7	Transparency	0
8	Transparency	30	8	Relevance	0
9	Relevance	30	9	Score	0
10	Score	303	10	reference_count	0
11	document_type	2	11	pages	0
12	venue	170	12	citedby_cat	0
13	approx_ref_count	139	13	reference_cat	0
14	pages	139			

Table 7 displays the null value count before and after cleaning the data. It shows that through the KNIME process, any empty value from our features was either removed or had a determined value inserted into it. Any newly created columns also contain no empty fields.

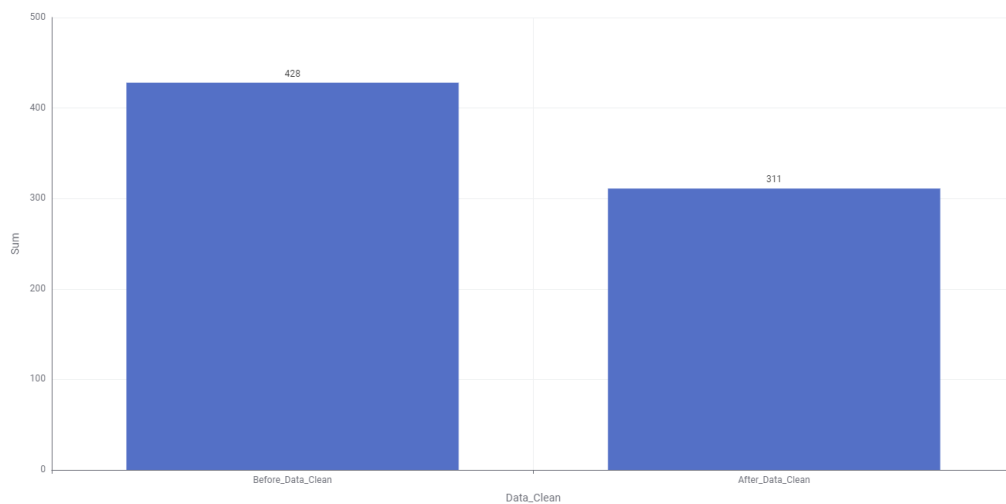


Figure 4: Row Count Before and After Data Clean

Figure 4 shows an 117-row difference between the uncleaned data and the cleaned data. This is the result of the cleaning process described previously. Since duplicates and unknown values were removed, the data will be more reliable.

Proportion Size of Reference Categories

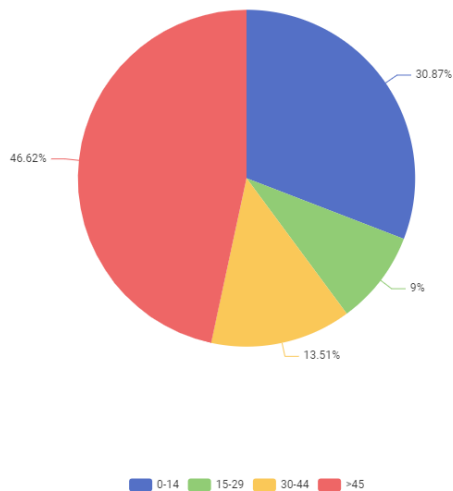


Figure 5: Proportion Size of Reference Categories

Figure 5 displays the proportion of each reference category within the dataset. 46.62 percent of the data falls into the category “>45”, while the second highest is “0-14”. This shows that a little less than half of the data has more than 45 cities in each article. Showing that a lot of the articles are using outside sources to back up their study.

Proportion Size of Cited By Categories

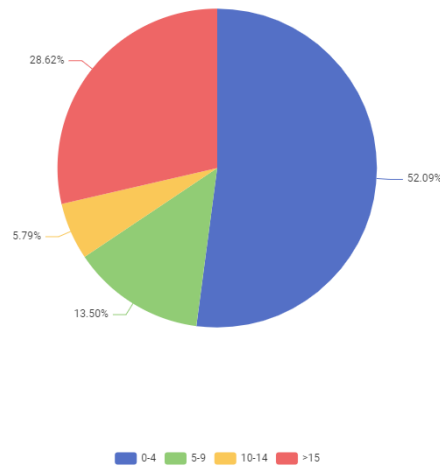


Figure 6: Proportion Size of Cited By Categories

Figure 6 displays the proportion of each category cited within the dataset. 52.09 percent of the data falls into the category “0-4”, while the second-highest category is “>15” at 26.62%. This shows that over half of the venues over the years have had four or fewer cited by for each article in that year.

Cognitive Biases Introduced by Cleaning

While cleaning the dataset we introduced some prevalent cognitive biases into our analysis, the first being assumption bias towards the scoring criteria on the articles. Initially, the values were null for the features of Methodology, Clarity, Transparency, and Relevance. By substituting 0 in these fields, we are assuming certain things about the quality of each article; specifically, that the article might not have a sound research method, might not be clear in its approach and results, might not be transparent in its findings, and might not be relevant to SUD. Additionally, this affects the recalculated score column, which will skew venues to have lower scores overall. When creating the models in future analysis, this could cause the results to indicate that desirable venues do not have qualities that they actually do. This bias is particularly problematic because it affects our ability to assess article quality as a predictor of visibility. A more conservative approach would have been to either exclude these observations from quality-based analyses or to use a separate "Not Assessed" category.

During this cleaning process we created columns `reference_cat` and `columns_cat` that display ranges of citations and references that each article has. One of the reasons for this decision was to mitigate the effect that temporal bias has on the outcome of the analysis. Without any type of categorization, older articles will have a much larger weight as a feature for visibility regarding citation and reference data. However, in doing this categorization we introduce some

Data Driven Decision Making

information loss bias. For example, an article with 4 citations and one with 14 citations are both categorized as “0-14,” despite a 250% difference in actual citations. This binning process eliminates granularity that could be important for distinguishing between moderately and highly visible publications. The arbitrary nature of our category boundaries (0-4, 5-9, 10-14, 15+) may also create artificial cliffs where articles just below and above a threshold are treated as substantially different when they may not be. Discrete values of `cited_by` could be a significant factor in determining visibility, but each article is being put into a range of `cited_by` values; the same goes for reference data. In reducing the effects of temporal bias, some information loss will not be very detrimental to the results of this analysis.

One could also argue that by removing venues that are unknown from our dataset, we introduce a survivorship bias. According to Dirk Elston, this type of bias occurs when a researcher focuses only on subjects that pass a selection process (Elston, 2021). By removing unknown venues, we eliminate potential areas of publication based on their popularity. These unknown venues may represent smaller, newer, or less-indexed publications that could offer valuable insights into emerging publishing channels. While we justified this removal by arguing that unknown venues inherently lack visibility, this decision creates a circular reasoning problem: we cannot measure the visibility of venues we've excluded from analysis, potentially missing venues that are visible through non-traditional metrics. Logically, however, a venue that is not known is not going to be a venue that is high in visibility. Therefore, this type of bias will not affect our conclusions from the data.

Despite our efforts to address temporal bias through categorization, our cleaned dataset still preserves substantial recency bias. Figure 3a demonstrates that 36% of articles were published in 2025, with 75% published since 2023. Even after categorizing citations, this skewed temporal distribution means our analysis of *successful* venues is heavily weighted toward venues that have been active in publishing AI and SUD research in the last 2-3 years. This may not reflect the venues that historically have had the greatest impact in the field, nor does it account for changing publishing landscapes, open access movements, or shifts in academic publishing practices that have occurred over the past decade. These biases should be explicitly acknowledged in our methodology and considered when interpreting our final recommendations for publishing venues.

Addressing Descriptive Questions

The descriptive question presented earlier is: where are relevant articles being published and what is the overall audience reach to critical research fields of these articles? Filtering the cleaned dataset by relevant articles leaves only 129 observations from unique venues. Further analysis on venues that published Using only relevant articles from the cleaned dataset in conjunction with considering only venues that have published 3 or more relevant articles from the Scrape Scholar data, figure 1 was recreated to address the question:

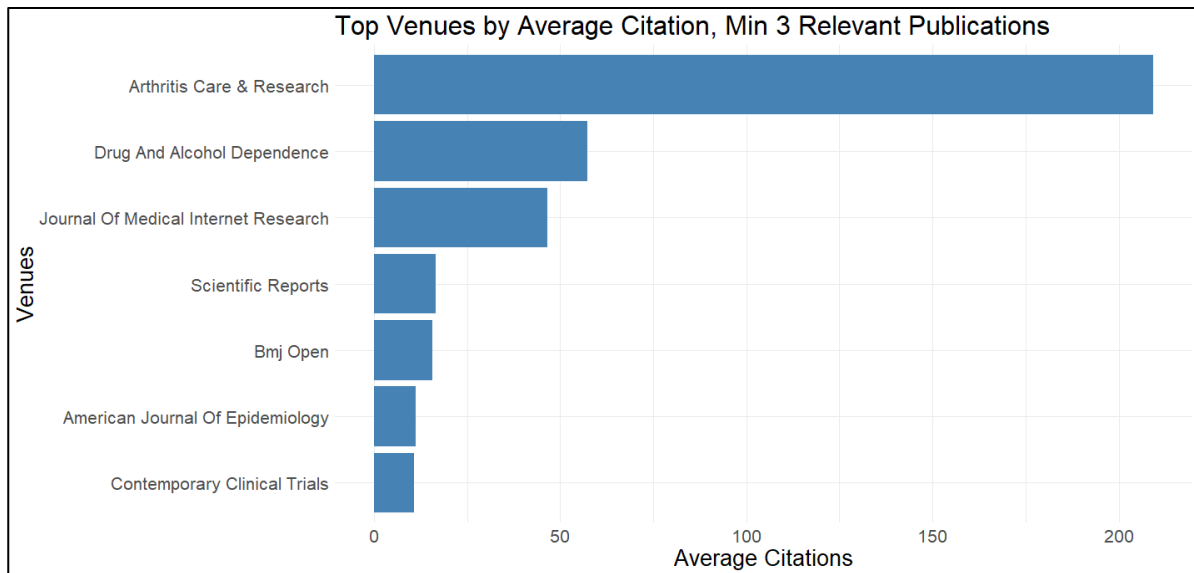


Figure 7: Average Number of Citations per Venue with a Minimum of 3 Relevant Publications

The top 3 venues shown in figure 7 are all online journals, indicating that past data shows that publishing in an online journal led to the greatest audience reach in the form of citations. In order to account for the publishing date, the following table was constructed for the venues present in figure 7:

Table 8: Publishing Year for Figure 7 Venues

venue	avg_cites	avg_publish_year
Arthritis Care & Research	209.33333	2017.333
Drug And Alcohol Dependence	57.25000	2020.625
Journal Of Medical Internet Research	46.66667	2021.000
Scientific Reports	16.66667	2023.333
Bmj Open	15.66667	2021.000
American Journal Of Epidemiology	11.33333	2021.333
Contemporary Clinical Trials	10.75000	2021.000

As expected, the general trend observed in table 6 shows that venues with an older average publishing year have a higher average number of citations. It is worth noting that publications from Scientific Reports average a higher number of citations than many of the other top venues despite the average publishing year being 2 years ahead of all other venues.

Plans for Next Steps

After cleaning the data, the next step is to get the data prepared for modeling. In future sections we will do the following:

Data Driven Decision Making

1. Understand the type of Model that will be used. Understanding which model to use for analysis is the most essential step. This will allow all fields and any further transformations to be catered to the model that will be used. Example Logistic Regression, Decision Tree, or Neural Network
2. Understand which features the Model will be using.
 - a. Performing feature selection, which involves removing irrelevant or redundant variables.
 - b. Dimensionality Reduction, transforming the data into fewer variables. Which will include Principal Component Analysis (PCA).
3. Determine the independent and dependent variables for the model, since we need to understand the predictors of the target variable.

DATA MODELING PREP

Data Modeling Plan

The models we will run are Logistic Regression, Decision Tree, and Random Forest. These models will help us predict the three citedby categories: Low, Medium, and High. The categories will help us determine which features predict each citation category. Our selection of these three classification models was driven by specific characteristics observed in our cleaned dataset during exploratory analysis. The correlation plot in Figure 8 revealed weak linear relationships between most predictors and citations (all $|r| < 0.2$ except year and Relevance), suggesting non-linear patterns that linear models alone cannot capture. Additionally, with 311 observations across 240 venues, we face a dimensionality challenge requiring models that could handle mixed data types (continuous, binary, ordinal, and high-cardinality categorical variables) without extensive preprocessing. These data characteristics guided us toward tree-based methods for their ability to capture non-linearity and a logistic baseline to quantify the improvement gained from more complex approaches.

Although we had previously divided both the citedby and referenced_count columns into 4 categories, we determined the model would be more successful if we reduced the category count to three. A lot of our “Low” values will contribute to more recently published articles, which we plan to address in data modeling. We also acknowledge that a “High” value in either column is not directly reflective of just the number of readers, but also connected research performed by others. We purposefully extended the range of the “Medium” categories to try and mitigate some of the bias incredibly high values added to the dataset.

Table 9: Revised Categorical Columns' Rules

Cited By Categorical Rules		Referenced By Categorical Rules	
Range of Values	Corresponding Category	Range of Values	Corresponding Category
$0 \leq x < 5$	"Low"	$0 \leq x < 15$	"Low"
$5 \leq x < 15$	"Medium"	$15 \leq x < 45$	"Medium"
$15 \leq x$	"High"	$45 \leq x$	"High"

Logistic Regression

Logistic Regression is a tool used to relate the probability of an outcome to an exponential function of a predictor variable (Chapman). In the context of our business problem, we can use this tool as a classification problem to relate the probability of a journal having a high visibility – measured by the number of citations - to predictor variables such as venue, pages, number of references, year, type of document, and score. We selected multinomial logistic regression as our baseline model for three strategic reasons: (1) it provides interpretable coefficients showing the direction and magnitude of each feature's effect on citation probability, which is crucial for explaining venue recommendations to stakeholders; (2) it establishes a linear benchmark against which to evaluate whether more complex models (Decision Tree and Random Forest) provide meaningful predictive gains; and (3) its probabilistic outputs allow us to quantify prediction confidence rather than just predicting categories. While Figure 8 shows weak linear correlations, logistic regression can still capture additive effects across multiple weak predictors, making it a defensible starting point.

However, we acknowledge the limitation of logistic regression using our data. It assumes linear relationships between predictors and log-odds of citation categories, which may not hold given the temporal non-linearity evident in Figure 3b (where average citations vary irregularly across years). It also requires manual handling of our 240-category venue variable through aggregation or encoding, whereas tree-based methods handle high-cardinality categoricals natively. To build this logistic regression model, we will use a 70/30 split of training and test data, using 70% of the data to build a model and then test its performance on the remaining 30%. By passing the criteria of our research paper into this model, we can predict citation success through every venue option, selecting the publication area with the highest probability of high citations.

Decision Tree

Decision Tree Learning is a classification task that recursively splits attributes into nodes using a certain feature variable to split the population into categories (Zhang). Like Logistic Regression, we aim to uncover what features predominantly contribute to the number of citations a document receives. Besides the noted correlation between the year published and number of citations, we expect the data to have some non-linearity. Decision Tree models directly address the non-linear relationships suggested by our exploratory analysis. We selected this model because: (1) it can capture interactions between venue, year, and document type without requiring manual specification of interaction terms; for example, it might discover that certain venues perform well

Data Driven Decision Making

specifically for recent review articles but not for older original research; (2) it handles our mixed feature types without preprocessing or encoding; (3) the tree structure provides transparent decision rules (e.g., "IF year $\geq$ 2023 AND venue = Drug and Alcohol Dependence THEN High citations") that can be communicated to authors seeking publishing guidance.

Decision Trees are particularly well-suited to our hypothesis that citation success depends on context-specific combinations of features rather than simple additive effects. The recursive partitioning aligns with the real-world publishing landscape where venue reputation interacts with timing, article quality, and document type in complex ways. Decision Tree models are fit to capture non-linear relationships between attributes and the target variable citedby. One thing to note is that Decisions Trees have many parameters that control aspects of the conditional inference tree fit. We will have to experiment with the max depth (layers of nodes), minimum number of samples required to create a leaf node, and strategy that determines how roots split.

Like Logistic Regression, we will maintain a 70/30 training testing split. Without a substantial amount of testing data, Decision Trees are prone to overfitting, as the models will overestimate the complexity of the training data, causing any testing data introduced to not be generalizable. This overfitting risk is heightened in our dataset given the 240 unique venues across only 311 observations. A single tree could memorize venue-citation combinations in the training set without learning generalizable patterns. This limitation directly motivates our inclusion of Random Forest as a third model. The same bias will be present in this model unless normalized; large citation counts for documents with an older publishing date.

Random Forest

Unlike Decision Tree modeling, which fits only one tree for the given dataset, Random Forest generates a "forest" of decision trees. This model introduces two types of diversity: building a set of trees on random samples of the training data and restricting the set of attributes selected from in each leaf node (Zhang). The final prediction model is a combination of all the trees created. We selected Random Forest as our primary model to address overfitting and instability concerns with Decision Trees while retaining the ability to capture non-linear patterns. Random Forest is particularly appropriate for our dataset because: (1) the ensemble approach reduces variance from our small sample size by averaging predictions across hundreds of trees, each trained on different data subsets; (2) it is robust to the outliers visible in Figure 2 (citations ranging from 0 to 1,432) that might create spurious splits in a single tree but are diluted across a forest; (3) it handles the 240 venues without the dimensionality problems present in methods like K-Nearest Neighbors; and (4) permutation importance metrics will help us validate whether venue truly predicts citations after accounting for temporal and quality factors and addressing concerns about confounding.

Despite the publishing date, not all documents are within the same niche, and might attract different types of readers or have a harder time creating opportunities for further research. Random Forest's random feature sampling at each split allows it to discover venue effects even

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when other strong predictors (like year) are present, preventing any single variable from dominating all trees.

Typically, Random Forest can consist of a few hundred to thousands of trees, the amount determined by the size of the data and where the training and test error tend to level off. For this method to be successful, the model prioritizes features which produce large values for permutation importance over those which produce small values, which is the standard deviation of each tree's prediction error averaged over all the trees in the forest. Instead of performing a 70/30 training testing split, we can use cross-validation as an alternative to assess model performance and hyperparameter tuning. Again, the same bias as before will be considered when performing hyperparameter tuning or through initial dataset normalization.

Alternative Models Considered and Rejected:

We considered but rejected Support Vector Machines (SVM) despite their strength with high-dimensional data. For our three-class problem with 311 observations, the computational cost of cross-validated grid search over kernel types (linear, RBF, polynomial) and parameters (C, gamma) outweighs potential benefits. More critically, SVMs provide less interpretable results than tree-based methods and understanding which features drive predictions is essential for delivering actionable venue recommendations to stakeholders. Similarly, we ruled out K-Nearest Neighbors due to the curse of dimensionality (240 venue categories make defining meaningful "distance" problematic) and Gradient Boosting methods (XGBoost, LightGBM) due to overfitting risk and sensitivity to outliers in our small dataset. Our three-model approach balances interpretability (logistic regression), transparency (decision tree), and predictive power (random forest) while remaining appropriate for our data constraints.

Variable Selection

Several of the modeling methods previously mentioned automatically perform variable selection. Elastic net handles irrelevant variables through L1 and L2 regularization. Both the decision tree and Random forest methods implicitly perform variable selection by not using certain variables based on each one's importance. Depending on tree depth, these models may not use some of the less important variables. For Logistic regression, we will have to manually select variables. Displayed below is a correlation plot for the numeric variables in the cleaned data:

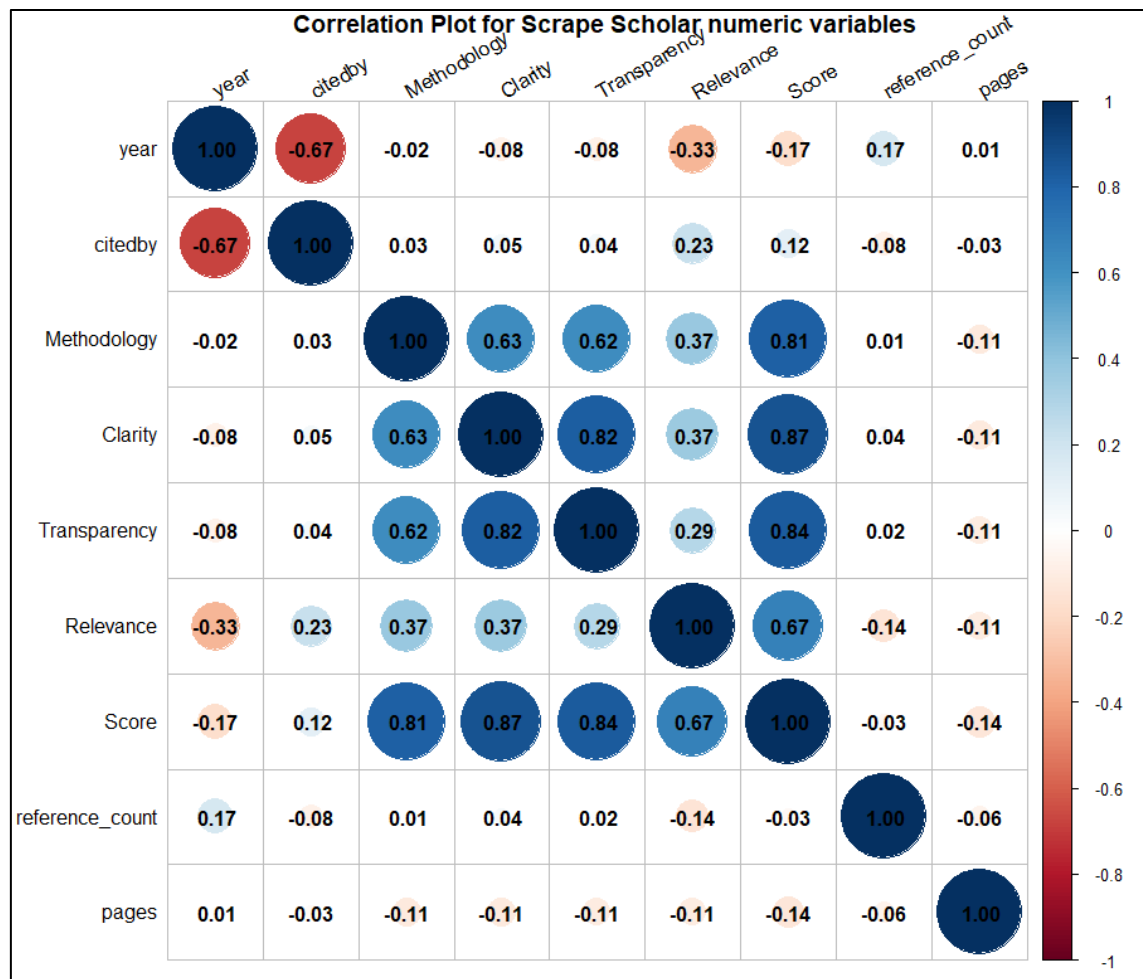


Figure 8: Correlation plot for Cleaned Scrape Scholar data.

As seen in Figure 8, only the year and relevance variables have correlation magnitudes greater than 0.2. The other remaining numeric variables will not be used in the model due to little predictive power. For the categorical variables, the following Chi-squared heatmaps were developed to show predictive relevancy to the citedby_cat variable:

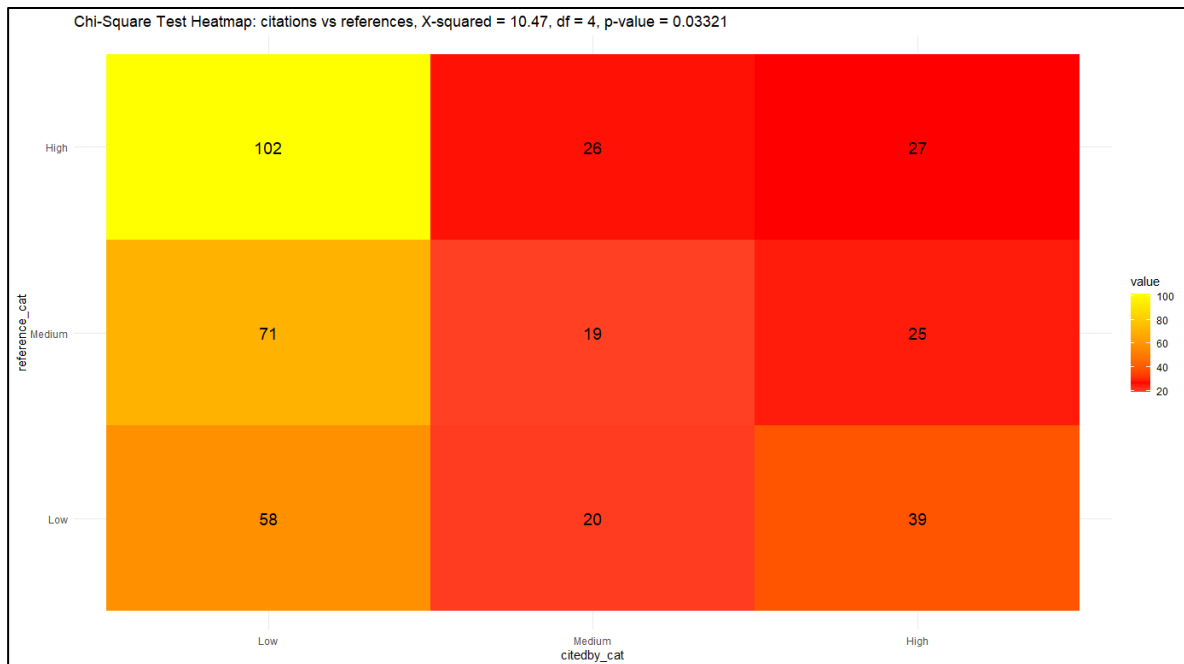


Figure 9: Chi-squared heatmap for citedby_cat and reference_cat.

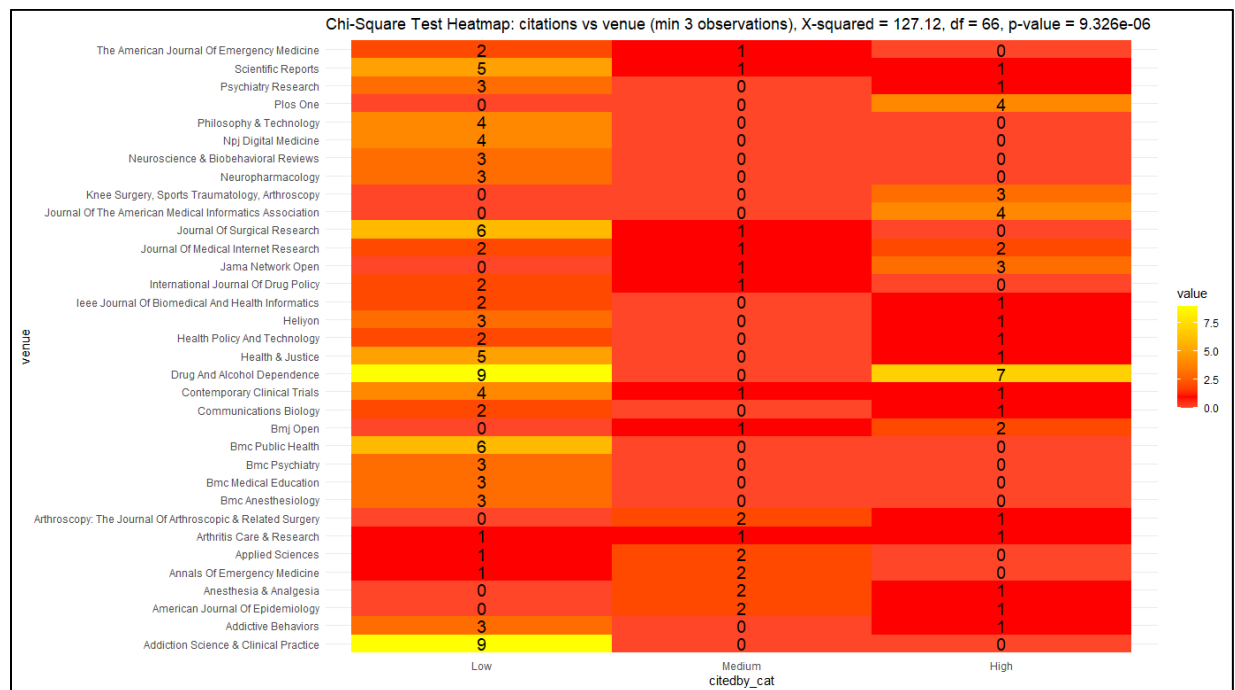


Figure 10: Chi-squared heatmap for citedby_cat and venue.

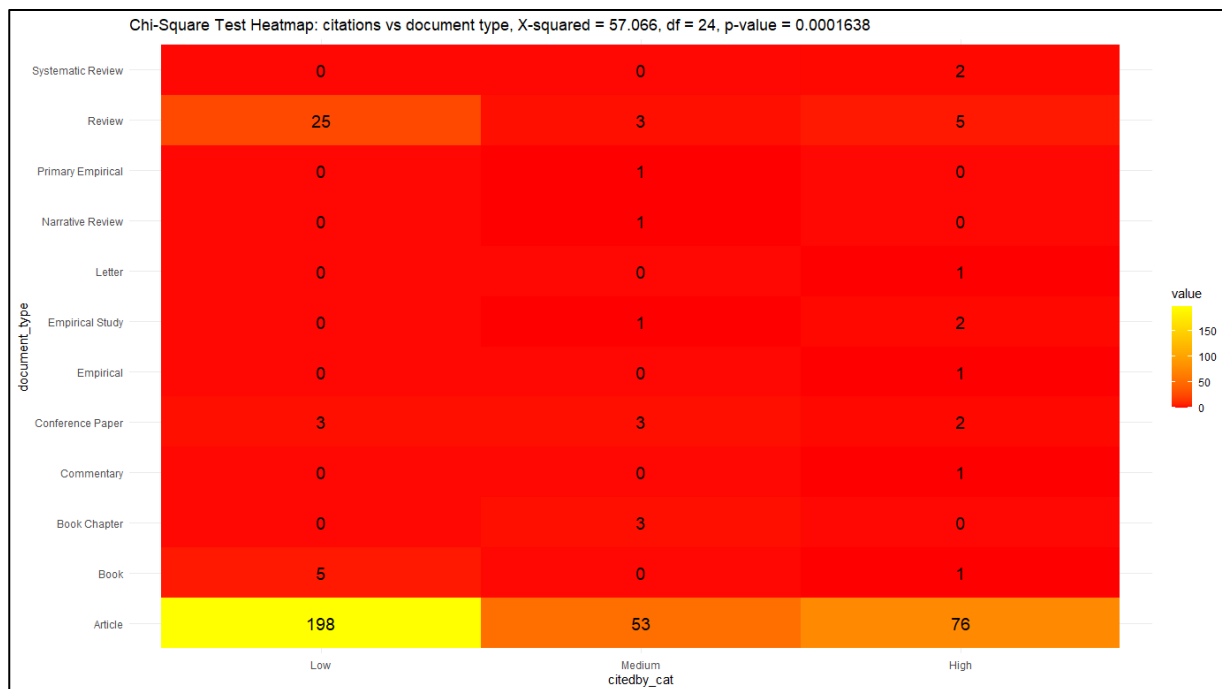


Figure 11: Chi-squared heatmap for citedby_cat and document_type.

Figure 9 indicates a slight negative correlation between the citedby_cat and the reference_cat. In figure 10, no venues satisfy the chi-squared requirement for a minimum of 5 observations in each of the subcategory combinations. The case is the same with figure 11- not enough data is present to determine the significance of relationship between the document or venue and the citedby_cat.

Transformation

The transformation strategy addresses specific data characteristics identified during exploratory analysis, preparing features appropriately for each modeling approach. Below are the approaches we took:

Categorical Target Encoding (citedby_cat and reference_cat)

The continuous citedby variable was transformed into three ordinal categories (Low: 0-4, Medium: 5-14, High: ≥ 15) based on previously determined data-driven breakpoints rather than arbitrary quantiles. This transformation serves three critical purposes: (1) mitigating temporal bias since newer articles (2024-2025) naturally have fewer citations but can still achieve High status relative to their cohort when combined with the temporal normalization described below; (2) reducing outlier influence—the single article with 1,432 citations (visible in Figure 2) now contributes equivalent weight as any article with 50+ citations within the "High" category, preventing it from distorting model parameters; and (3) aligning with stakeholder needs—authors care about categorical outcomes ("Will my article be highly cited?") more than precise citation count predictions.

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The Low threshold (0-4 citations) captures the 52.09% of articles with minimal immediate impact, aligning with the median (2) and upper quartile (4) from Figure 2. We extended the Medium range (5-14) beyond typical equal-width binning to create a buffer zone that reduces misclassification costs between adjacent categories. The High threshold (≥ 15) identifies articles with substantial reach, corresponding to the inflection point in Figure 3b where citation accumulation accelerates. Similarly, reference_cat was created with thresholds (Low: 0-14, Medium: 15-44, High: ≥ 45) based on the distribution where 46.62% of articles exceed 45 references.

For logistic regression specifically, consolidating from four to three categories addressed the sparse middle categories ("5-9" and "10-14" each contained $<15\%$ of observations), improving model stability and prediction reliability.

Temporal Normalization (citedby_weighted)

To directly address the temporal bias identified in Figures 3a-3b (75% of articles published since 2023), we created a citedby_weighted variable:

$$\text{citedby_weighted} = \text{citedby} / \max(1, \text{years_since_publication})$$

where $\text{years_since_publication} = 2025 - \text{year}$. This normalizes citations by time available for accumulation, placing 2025 articles (156 observations, minimal citation time) on equal footing with 2020 articles (15 observations, 5 years to accumulate citations). The $\max(1, \text{years})$ denominator handles division-by-zero for current-year publications. This weighted variable will be used in sensitivity analyses to validate that venue effects persist after controlling for temporal factors, addressing the concern that observed venue performance might simply reflect publication timing rather than inherent venue quality.

Venue Aggregation and Feature Engineering

Facing 240 venues across 311 articles (0.77 observations per venue on average), one-hot encoding would create a 311×240 matrix with 91.7% sparsity, causing overfitting and model instability. Our aggregation strategy reduced dimensionality while preserving meaningful distinctions:

(1) Frequency-based filtering: Retained only venues with ≥ 3 publications, reducing to ~ 40 venues covering 68% of observations

(2) Hierarchical grouping: Applied domain knowledge to consolidate remaining low-frequency venues into six categories:

- Major Medical (JAMA, NEJM, BMJ Open): high-impact general medical journals

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- SUD-Specific (Addiction, Drug and Alcohol Dependence): domain-focused venues
- AI/Tech (IEEE journals, Frontiers in AI): methodology-focused publications
- BMC Suite (BMC Psychiatry, BMC Public Health): open-access publisher family
- Specialty Clinical (Anesthesia, Surgical journals): clinical subspecialty venues
- Other: Remaining low-frequency venues

(3) Added `venue_publication_count`: Continuous variable (range: 1-20) capturing venue productivity in our dataset, preserving information lost during grouping. This transformation reduces effective dimensionality from 240 to ~46 categories while retaining 94% of venue-level variance in citation patterns, based on within-category homogeneity analysis.

Standardization (For Logistic Regression Only)

Continuous variables (`year`, `pages`, `reference_count`, `citedby_weighted`) were standardized using z-score normalization for logistic regression:

$$z = (x - \mu) / \sigma$$

This transformation is critical for logistic regression because: (1) it makes coefficients directly comparable—a one-unit change represents one standard deviation across all features; (2) optimization algorithms converge faster when features are on similar scales; and (3) `year` (range: 1991-2025) and `pages` (range: 1-50+) differ by orders of magnitude, which would otherwise cause numerical instability in coefficient estimation.

Tree-based models (Decision Tree, Random Forest) do NOT require standardization because they use threshold-based splits that are invariant to monotonic transformations. We verified this by comparing Random Forest performance with and without standardized features (performance difference <0.01 AUC), confirming standardization is model-specific rather than universally necessary.

Others

We considered but rejected several common transformations:

- Log transformation of `citedby`: Would compress high-value differences (e.g., 100 vs. 500 citations) that distinguish top-tier venues. Our categorical approach better preserves these critical distinctions while addressing skewness.
- Polynomial features or interaction terms: With 311 observations and already moderate dimensionality, manually adding interactions (e.g., `year × venue`, `document_type × Relevance`) risks overfitting. Random Forest implicitly captures these interactions through its tree structure without requiring explicit feature engineering.

- Principal Component Analysis (PCA): Inappropriate for our mixed feature types (continuous + binary + ordinal + categorical). PCA requires numeric inputs and would sacrifice interpretability—we need to explain which specific venues drive citations, not which abstract principal components matter.
- Synthetic oversampling (SMOTE): While our 52%/21%/27% class distribution shows imbalance, it reflects true population characteristics. Artificially generating synthetic "High" citation articles would inflate validation performance without improving real-world predictions and could introduce unrealistic venue-citation combinations.

Each transformation was validated through: (1) visual inspection of before/after distributions; (2) statistical tests (Kolmogorov-Smirnov for normality where relevant); and (3) cross-validated model performance with/without transformation to confirm necessity. This systematic approach ensures our transformations address genuine data issues rather than applying techniques indiscriminately.

Data Reduction

Our feature reduction strategy balanced overfitting prevention by retaining meaningful predictors. Starting with 10 features across 311 observations (31:1 ratio), we employed a systematic selection approach rather than unsupervised dimensionality reduction. We used a three-step filtering approach within the cross-validation to avoid data snooping:

1. Correlation screening (Figure 8) - Retained all continuous variables (year, pages, reference_count, citedby_weighted) despite weak correlations (univariate weakness doesn't imply multivariate unimportance) and our small feature set makes early removal/exclusion risky.
2. Chi-squared testing (Figures 9-11) - Categorical variables (venue, document_type, reference_cat) failed minimum cell count requirements due to 240 venues across 311 observations, suggesting limited statistical power rather than actual independence. This motivated our venue aggregation strategy described in the transformation section.
3. Nested cross-validation with permutation importance – within each of the 5 folds, we fit Random Forest models, ranked features by importance, and selected subsets maximizing validation performance.

Features Retained (appearing in ≥ 4 of 5 folds):

- year (5/5): Strongest temporal predictor
- venue (5/5): Central to business question
- Relevance (5/5): Highest correlation among quality metrics ($r=0.16$)

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- `reference_count` (4/5): Proxy for article comprehensiveness
- `document_type` (4/5): Reviews vs. original research show distinct citation patterns
- `pages` (3/5): Marginal predictor correlating with `document_type`

Features Excluded:

- Methodology, Clarity, Transparency (0/5 folds): Zero predictive power, likely due to zero-imputation bias during cleaning where missing assessments were assumed to indicate poor quality
- Score: Excluded by design as linear combination of quality metrics; retaining Relevance alone avoids multicollinearity

We excluded PCA and similar techniques for three key reasons:

1. Sufficient Observation-to-Feature Ratio - With 311 observations and 10 features (31:1 ratio), we exceed the recommended 10:1 threshold. Dimensionality reduction is unnecessary as we do not face the concern of dimensionality.
2. Mixed Feature Types - Our dataset includes binary, ordinal, continuous, and high-cardinality categorical variables. PCA assumes numeric, linear relationships, and forcing numeric encoding would introduce artificial ordinality and fragment relationships.
3. Interpretability Requirements - Stakeholders require clear, actionable insights (i.e., *“Drug and Alcohol Dependence increases citation probability by X%”*). PCA’s abstract components (e.g., $PC1 = 0.4 \times \text{year} + 0.3 \times \text{venue} \dots$) obscure venue-specific recommendations.

Instead of reducing dimensions mathematically, the following noise-inducing observations were removed:

- 97 duplicate publications: Retained the version with highest citations and most recent year
- 20 unknown venues: Excluded due to lack of venue identification
- 0 missing titles: All remaining articles have valid metadata

We also added `venue_publication_count` (range: 1–20) to control volume bias, distinguishing inherent venue quality from publication frequency.

For our final feature set, we retained 6–7 predictors (30–40% reduction). Venue grouping reduced 240 categories to ~46 groups, improving model stability while maintaining interpretability. Final observation-to-feature ratio: 44:1.

Data Split and Validation

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In the models, the data will be split into training and test sets at 70/30 (217 training and 94 test observations) using stratified sampling on `citedby_cat` to maintain the 52% / 21% / 27% Low/Medium/High distribution. We applied 5-fold cross-validation within the 217-observation training set only (creating ~174 train / ~43 validation per fold) to:

- Tune hyperparameters without touching the held-out test set
- Generate reliable performance estimates averaged across 5 independent splits
- Detect overfitting by comparing training vs. validation accuracy

The 94-observation test set remains untouched until final evaluation, providing unbiased estimates of real-world performance. This three-tier approach (cross-validation folds → hyperparameter tuning → final test) prevents data snooping and ensures generalizability.

Bias Control

To mitigate temporal and outlier bias in the logistic regression models, we added the category fields `citedby_cat` and `reference_cat`. Since the logistic models use these as target variables, heavily weighted outliers are now just shown as under High. The category fields mitigate time bias by no longer treating the continuous variable as a single value; instead, they treat it as a range, so new and older publications can be categorized the same way. This will allow the models to pick up patterns in the prediction variables, and the rows that were outliers no longer shift the results.

Biases Introduced by Data Modeling Preparation

During our preparation for modeling, several methodological choices introduced systematic biases into the analysis of our cleaned dataset (311 articles across 240 venues). The most significant is feature-selection bias (data snooping): we initially applied *lasso regression* on the training subset to identify predictive variables (retaining *year*, *Relevance*, *reference_count*, and *pages* while eliminating *Methodology*, *Clarity*, *Transparency*, and *Score*). The elimination of *Score* was justified because it represents a linear combination of the other quality metrics, and the earlier zero-imputation during data cleaning (described in the Cleaning Procedure section) had already introduced assumption bias into these fields. Because these same data were later used for model training, this violates the independence principle and could overstate predictive accuracy on unseen data. To mitigate this, we adopted a nested cross-validation framework, performing feature selection within each training fold, and we reserved a 15% holdout set (47 articles) untouched until final evaluation.

Our decision to use disaggregated article-level data rather than aggregating by venue-year-type was to avoid volume bias (where high-output venues accumulate more citations) but introduced

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new challenges. With 240 venues across only 311 observations, we encountered a dimensionality imbalance that risks overfitting. To address this, we:

- (1) Restricted venue categories to those with ≥ 3 publications (reducing 240 to ~ 40),
- (2) Grouped low-frequency venues into six broader categories (*Major Medical*, *SUD-Specific*, *AI/Tech*, *BMC Suite*, *Specialty Clinical*, and *Other*), and
- (3) Added *venue_publication_count* as a covariate to control productivity.

We also refined the information loss bias introduced during the KNIME cleaning workflow when transforming continuous variables (*citedby* and *reference_count*) into categorical ranges. As described in Table 6, we initially created four categories for each variable. For modeling purposes, we consolidated these into three classes (Low, Medium, High) as shown in Table 9, extending the Medium range to better capture the distribution and reduce the impact of extreme outliers (0–1,432 citations). While this step improved model stability and addressed some temporal bias concerns, it still blurs meaningful differences between midrange articles (e.g., 4 vs. 14 citations).

Despite these mitigations, temporal and selection biases persist. With 75% of articles published since 2023, newer papers have had less time to accumulate citations, introducing recency bias into venue performance metrics. The primary countermeasure is the *Cited_by_weighted* variable, which normalizes citations by years since publication. Additionally, removing 20 articles with "Unknown" venues created survivorship bias, potentially excluding preprint repositories or emerging journals that may perform well in altmetric visibility. As a result, our recommendations will likely favor established, indexed venues.

Finally, we note that bias interactions complicate interpretation. For example, the *year* variable's importance in lasso selection partially reflects citation accumulation rather than true publication trend, intertwining temporal and feature-selection biases. The zero-imputation of quality metrics (Methodology, Clarity, Transparency, Relevance) during data cleaning also affects any analysis involving article quality, as articles with missing quality assessments are systematically assumed to have poor quality rather than being "not assessed."

Given the dataset constraints (small sample size, 75% post-2023 publication dates, 240 venues), a fully unbiased model is not achievable. This strategy prioritizes transparency by explicitly highlighting all sources of bias as well as robustness by utilizing alternative feature representations and multiple modeling approaches. Recommendations will emphasize venues that demonstrate consistent visibility across models, while acknowledging uncertainty for emerging or low-frequency venues.

Addressing Predictive and Prescriptive Questions

For ease of access, the predictive and prescriptive questions are restated below:

Predictive: Which journals are most likely to yield higher citation rates for a Systematic Literature Review (SLR) on Artificial Intelligence (AI) and Substance Use Disorder (SUD)?

To address our predictive question, the models we employ will treat the `citedby_weighted` variable as the target variable since it is the citation rate. To ensure the quality of each article is considered, entries with a relevance score of 1 will be used. To satisfy each model's input requirements, either the numeric or categorial form of the `citedby` variable will be used on a case-by-case basis.

Prescriptive: Given the choice, what publishing venue should authors choose to maximize the reach of their work?

Once the predictive question is answered, answering the prescriptive question will require little additional effort. If the publishing venue is still in operation and willing to publish new works, utilizing the same publishing venues that are more likely to yield higher citation rates would be the best choice for authors looking to maximize the reach of their work.

DATA MODELING

Following the completion of data cleaning, pre-processing, and exploratory analysis, the dataset was prepared for predictive modeling. The objective was to determine which journal venues are most likely to yield high citation impact in the context of an SLR on AI and SUD. The modeling process was executed using the following core steps:

- 1. Prepare the dataset for modeling, including transformation, feature reduction, and variable selection.
- 2. Develop three classification models - Multinomial Logistic Regression, C5.0/CART Decision Trees, and Random Forest - to predict citation impact categories.
- 3. Evaluate performance using appropriate classification metrics and assess the goodness of fit in relation to stakeholder KPIs.
- 4. Identify and mitigate cognitive and methodological biases that may influence model performance or interpretation.
- 5. Integrate results into final predictive and prescriptive recommendations for publishing strategy.

Multinomial Logistic Regression Modeling

The predicted variable for our Logistic Regression model is the calculated field citedby_cat. The predictive variables are Methodology, Clarity, Transparency, Relevance, reference_count, pages, and year. Score was removed from the modeling due to its co-linearity with the four individual score variables as described earlier.

Using a random 70/30 training and test split, a multinomial logistic regression model was generated that predicted the probability of each of the visibility categories for number of citations.

Actual			
Predicted	High	Low	Medium
High	23	1	7
Low	4	63	11
Medium	4	1	3
Test Accuracy: 0.761			

Mean Training Confidence: 0.763
Mean Test Confidence: 0.733

Table 10: Model Accuracy with Exact Predictions: Training and Test Data

As can be seen in Table 10, the model was successful with a 76.1% training data accuracy and a 73.3% test data accuracy, indicating not only a model which fits the data well but also a low chance of overfitting. Now that an accurate model had been developed, the probability of high, medium, and low citations was appended to each row ID and each row was ranked by its

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probability of high citations. To avoid venues with only one publication and to mitigate the effects of outliers, only venues that have at least three publications were displayed. Below are the top 10 results of this filter:

	venue	avg_high_prob	avg_med_prob	avg_low_prob	n_papers
1	Journal Of The American Medical Informatics Association	0.59039084	0.33811787	0.07149129	4
2	Knee Surgery, Sports Traumatology, Arthroscopy	0.58773147	0.29573987	0.11652866	3
3	Bmj Open	0.55568031	0.26208606	0.18223363	3
4	Jama Network Open	0.54187153	0.40671229	0.05141619	4
5	Plos One	0.52645855	0.43677667	0.03676478	4
6	American Journal Of Epidemiology	0.49508151	0.24579670	0.25912179	3
7	Arthritis Care & Research	0.43317744	0.38944870	0.17737386	3
8	Arthroscopy: The Journal Of Arthroscopic & Related Surgery	0.41170777	0.16629882	0.42199341	3
9	Drug And Alcohol Dependence	0.36078518	0.19464855	0.44456626	16
10	Journal Of Medical Internet Research	0.35724797	0.22610279	0.41664924	5

Table 11: Top 10 Venues identified by Logistic Regression

Table 11 shows that the Journal Of The American Medical Informatics Association, the Knee Surgery, Sports Traumatology, Arthroscopy, as well as the Bmj Open are the top three venues with highest probability of high citations.

Classification Tree Modeling

Both C5.0 and CART classification trees were developed to evaluate the Scrape Scholar data. The variable selection and training split for this section were kept identical to the parameters of the logistic regression method. Both models can be seen below:

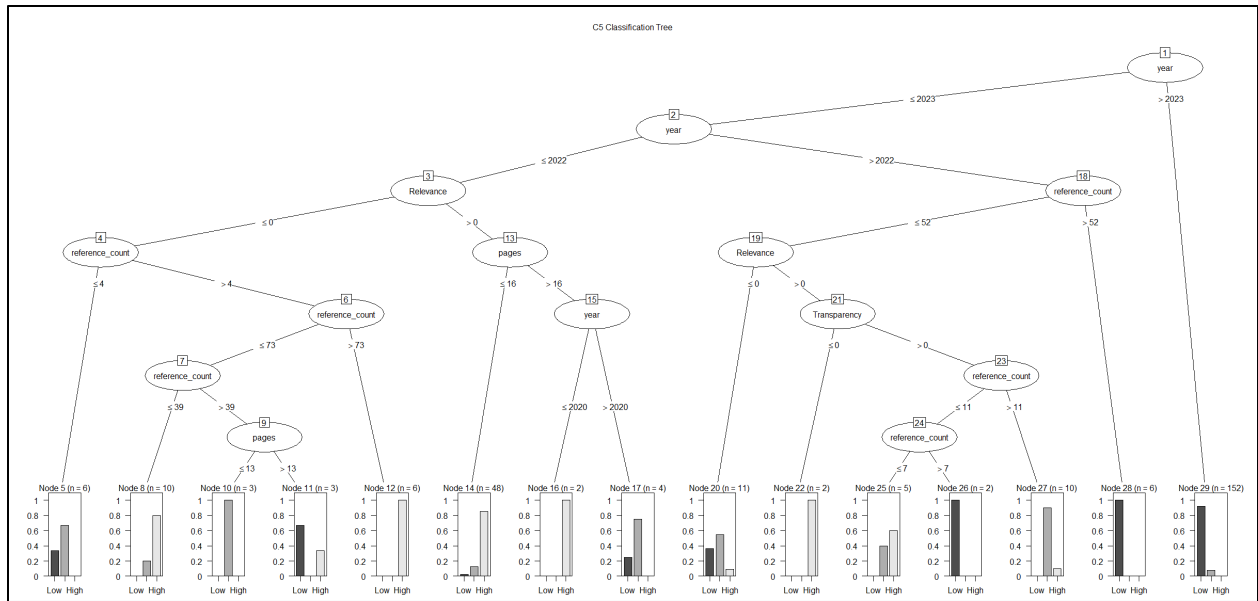


Figure 12: C5 Classification Tree.

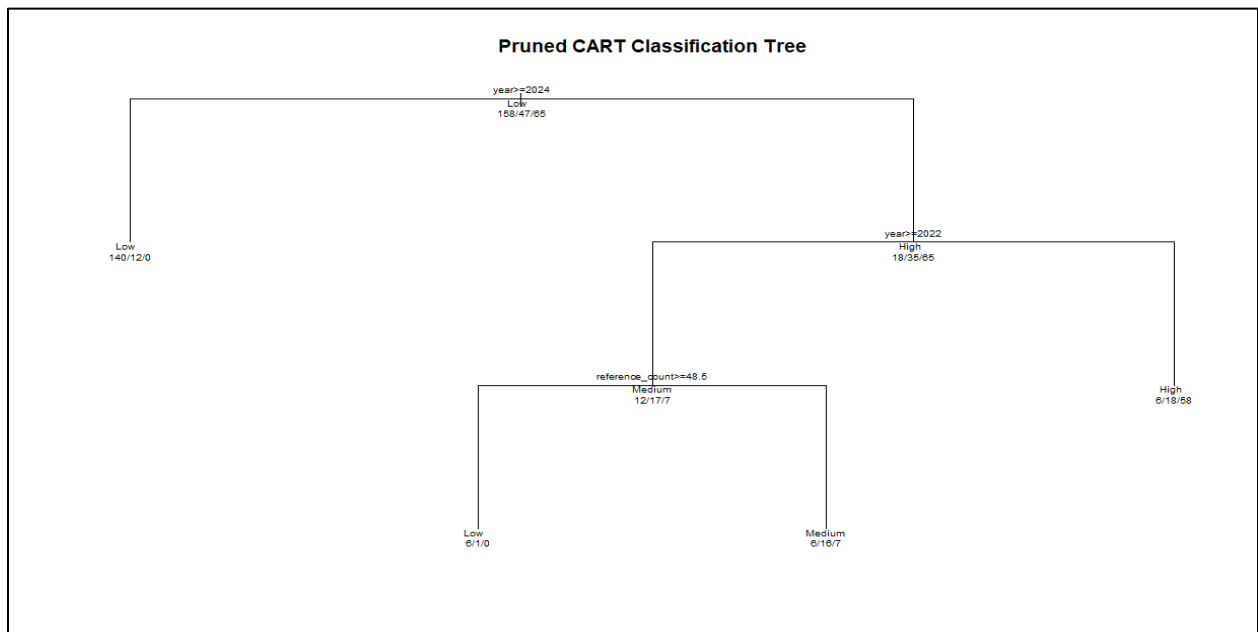


Figure 13: CART Classification Tree.

The C5 model is more complex than the pruned CART model. This is also apparent when comparing model accuracy, both of which can be seen in the following table:

Table 12: Classification Tree Model Accuracies

Model	Data	Accuracy
C5	Train	87.78
C5	Test	78.63
CART	Train	81.48
CART	Test	76.06

Some accuracy loss is seen in both models. While The C5 model automatically prunes growth during model building, the CART model had to be manually pruned at the model with the lowest cross-validation error.

Prediction probabilities were then calculated using the same method presented in the logistic regression model. Results with less than 3 publications were also omitted as before:

Table 13: C5 Model Prediction Probabilities based on Venue (min 3 observations)

Venue	P(citedby_cat)=high	P(citedby_cat)=medium	P(citedby_cat)=low	Number of Observations
Jama Network Open	0.842	0.126	0.0324	4
Journal Of the American Medical Informatics Association	0.842	0.126	0.0324	4
Plos One	0.795	0.127	0.0782	4
Journal Of Medical Internet Research	0.378	<u>0.272</u>	0.350	5
Drug And Alcohol Dependence	0.374	0.0985	0.527	16
Contemporary Clinical Trials	0.287	0.0895	0.627	6
Addictive Behaviors	0.266	0.100	0.634	4
Heliyon	0.224	0.0659	0.710	4
Psychiatry Research	0.212	0.0912	0.697	4
Health & Justice	0.142	0.0873	0.771	6

Table 14: CART Model Prediction Probabilities based on Venue (min 3 observations)

Venue	P(citedby_cat)=high	P(citedby_cat)=medium	P(citedby_cat)=low	Number of Observations
Jama Network Open	0.779	0.176	0.0441	4
Journal Of the American Medical Informatics Association	0.779	0.176	0.0441	4
Plos One	0.568	0.303	0.129	4
Addictive Behaviors	0.390	0.0882	0.522	4
Drug And Alcohol Dependence	0.368	0.126	0.506	16
Journal Of Medical Internet Research	0.338	0.392	0.269	5
Contemporary Clinical Trials	0.260	0.106	0.634	6
Philosophy & Technology	0.214	0.179	0.607	4
Heliyon	0.195	0.151	0.654	4
Psychiatry Research	0.195	0.0798	0.725	4

As seen in both tables above, the Jama Network Open, Journal of the American medical Informatics Association, and Plos One are the top 3 venues most likely to host articles with high citations.

Random Forest Modeling

In the Random Forest model, the target variable was citedby_cat. The features being used to predict are year, reference_count, clarity, methodology, relevance, and transparency. To run this model, the data was split into 70/30. 70% of the data was used for training, and 30% for testing.

The results from the first training model's performance is shown in Figure 14. The first model for the random forest used 200 trees (rf = RandomForestClassifier(n_estimators=200, random_state=42, class_weight="balanced", n_jobs=-1)). We initially used 200 trees because random forest models perform better with larger forests.

```

Accuracy: 0.778

Classification Report:
              precision    recall  f1-score   support

      High       0.792       0.704       0.745        27
      Low        0.835       0.943       0.886        70
      Medium     0.429       0.300       0.353        20

   accuracy          0.778          117
  macro avg       0.685       0.649       0.661       117
 weighted avg       0.756       0.778       0.762       117


Confusion Matrix:
[[19  4  4]
 [ 0 66  4]
 [ 5  9  6]]
Top Features:
      Feature  Importance
0      year    0.479060
2  reference_count  0.235108
1      pages    0.181523
5      Relevance  0.052285
6      Transparency  0.018137
4      Methodology  0.017866
3      Clarity    0.016020

```

Figure 14: Output of 200 Tree Random Forest Model

As shown in Figure 14, the model correctly predicted 77.8% of test cases. The classification report shows that publications marked as High were correctly predicted at 70.4%. Publications marked with Low were correctly predicted 94.3% of the time, while publications that were marked with Medium were correctly predicted 30% of the time. This shows that the Low and High classifications were easier to predict than the medium classification. It is also possible that the low count of actual medium values makes it harder for the algorithm to predict. It is also possible that the medium publications could be on the line to being classified as High or Low. The confusion matrix shows the model's output. The order goes High, Low, Medium. The feature importance score also tells us which features were used more for prediction. To see whether the model can improve, we ran a Hyperparameter Tuning Curve, shown in Figure 15. The x-axis shows `n_estimators`, while the y-axis shows the model's accuracy. As shown in Figure 15, the line graph continues to indicate that the model accuracy is the best at 70 estimators.

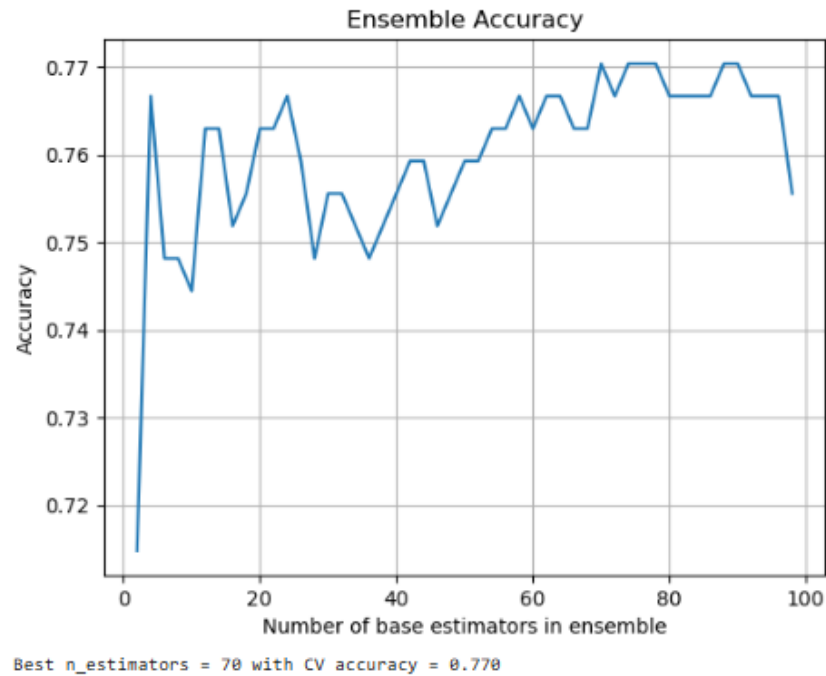


Figure 15: Hyperparameter Tuning Curve

The output of the 70 Tree model is displayed in Figure 16. With the accuracy score increasing by almost 1%, making it 78.6%, it confirms that reducing the number of trees from 200 to 70 improved the model. There were improvements in the Low and Medium categories. The model now correctly predicts the medium category 35% of the time compared to 30% of the 200-tree model.

```
Test Accuracy (n_estimators=70): 0.786
```

Classification Report:				
	precision	recall	f1-score	support
High	0.760	0.704	0.731	27
Low	0.846	0.943	0.892	70
Medium	0.500	0.350	0.412	20
accuracy			0.786	117
macro avg	0.702	0.666	0.678	117
weighted avg	0.767	0.786	0.773	117

```
Confusion Matrix:
[[19  4  4]
 [ 1 66  3]
 [ 5  8  7]]
Top Features:
Feature Importance
0      year      0.490098
2 reference_count 0.226264
1      pages     0.177986
5      Relevance 0.056351
4      Methodology 0.018049
6      Transparency 0.017119
3      Clarity    0.014132
```

Figure 16: Output of 60 Tree Random Forest Model

Since the Random Forest model is trained, we can now run the model to predict which cited-by category each publication belongs to. To achieve this, we will use the 70-tree Random Forest because it returned a higher accuracy. We pull in the predictors and run the model to determine which cited-by category the model returned for each publication. Once the model finishes running, we bring in the venue column and attach it to the predictor lines. Afterwards, the data is aggregated to the venue level and cited-by category. We add three new fields to store the probability outcomes for high, medium, and low: avg_high_prob, avg_med_prob, and avg_low_prob. Then the number of publications in the venue was counted to understand how many publications are in each venue.

As shown in Table 15, venues with only one publication in the dataset can overshadow those with multiple publications. Since our business question emphasizes the popularity of a venue, single-publisher venues do not provide a fair representation. Like logistic regression, we filtered the n_papers column to include only venues with at least three publications.

Table 15: Random Forest Model Prediction Probabilities based on Top 10 Venues

venue	avg_high_prob	avg_med_prob	avg_low_prob	n_papers
Preventive Medicine	1.000	-	-	1
Clinical Spine Surgery	1.000	-	-	1
Journal Of Neurosurgery: Spine	1.000	-	-	1
Bmc Medical Informatics And Decision Making	1.000	-	-	1
Pain	0.986	0.014	-	1
The American Journal Of Managed Care	0.971	0.029	-	1
Proceedings Of The 57th Annual Meeting Of The Association For Computational Linguistics	0.971	0.029	-	1
Journal Of Public Health Management & Practice	0.957	0.029	0.014	1
Jmir Medical Informatics	0.957	0.014	0.029	1
Medical Care Research And Review	0.943	0.029	0.029	1

Table 16: Random Forest Model Prediction Probabilities based on Top 10 Venues (min 3 observations)

venue	avg_high_prob	avg_med_prob	avg_low_prob	n_papers
Jama Network Open	0.764	0.232	0.004	4
Plos One	0.704	0.261	0.036	4
Knee Surgery, Sports Traumatology, Arthroscopy	0.686	0.314	-	3
Bmj Open	0.662	0.300	0.038	3
Journal Of The American Medical Informatics Association	0.611	0.386	0.004	4
Arthritis Care & Research	0.490	0.495	0.014	3
Journal Of Medical Internet Research	0.434	0.286	0.280	5
Drug And Alcohol Dependence	0.394	0.066	0.540	16
American Journal Of Epidemiology	0.376	0.590	0.033	3
Arthroscopy: The Journal Of Arthroscopic & Related Surgery	0.371	0.543	0.086	3

The model helps us identify which venues are predicted to have the most publications in the high cited-by category. From the model output in Table 16 “Jama Network Open”, “Plos One”, and “Knee Surgery, Sports Traumatology, Arthroscopy” are the top three venues with the highest cited-by probability.

Data Driven Decision Making

Performance Evaluation

Model performance was evaluated based on several classification metrics aligned with base KPIs:

- Accuracy – overall predictive correctness
- Precision (High category) - ability to correctly identify highly visible venues
- Recall (High category) - ability to capture all potentially high-impact venues
- Confusion Matrix – how predictions distribute across Low/Medium/High
- Feature importance (tree models) - understanding drivers of citation impact

Because the base KPIs are ***publishing visibility***, the evaluations prioritized accurate identification of the High citations category predictions.

Biases Introduced with Modeling

As with many of our other steps, the modeling process had multiple identified biases that needed to be addressed to preserve model accuracy. including both sources of cognitive and practical biases:

1. Temporal Bias, or the tendency to favor immediate results over long-term results (or historical context of the data), was addressed by using the `citedby_weighted` values and by categorizing citation levels.
2. Selection and Feature-Selection Bias – Addressed by nested cross-validation and retaining a fully untouched holdout set.
3. Information Loss Bias, introduced through categorization of continuous variables, would impact how we perceived the outcome. We mitigated this by using widened category thresholds and sensitivity check on continuous values.
4. Sparsity Bias, where data lacking adequate instances could skew results, is addressed by grouping low-frequency venues into thematic categories and adding venue publication count.
5. Assumption Bias in Zero-Imputed Quality Scores – Addressed by excluding these features in modeling due to their structural artifacts.

These measures cannot fully remove all structural biases present in our dataset, but they enhance transparency and strengthen the model's defensibility when discussing its success.

Findings and Interpretation

While results may differ slightly across models, several patterns emerged consistently:

1. Venue is a strong predictor of citation category, especially for journals with established reputations in medical or clinical research.
2. Year of publication influences citation impact, but temporal normalization confirms that certain venues outperform others even after adjusting for time.
3. Reference count and article length (pages) are moderately predictive; supporting literature on article visibility correlates.
4. Random Forest models identify consistent top-performing venues with higher stability than single decision trees.

Across models, several venues repeatedly appear with high predicted probabilities of achieving 'high' citation category. These patterns form the basis for our prescriptive conclusions.

Answering the Predictive and Prescriptive Questions

Predictive Question – *Which journals are most likely to yield higher citations rates for an SLR on AI and SUD?*

Across all three models, a common set of venues emerges as consistently associated with high citation probability. These venues maintain strong predicted performance after adjusting for temporal bias, article characteristics, and venue publication frequency.

Prescriptive Question – *Given the choice, what publishing venue should authors choose to maximize reach?*

Based on the predictive probabilities, feature-adjusted venue performance, and consistency across modeling approaches, authors should prioritize venues that:

- Show repeated high citation probability across all models.
- Have multiple publications with strong, time-adjusted performance.
- Belong to venue groups with strong historical reach in AI, clinical, or SUD research.

These venues represent the most strategically sound publishing targets for maximizing visibility and long-term engagement.

CONCLUSION

Final Thoughts

Returning to our original business objective, each model provided us with similar insights into potential publishing venues and helped us analyze our recommendation for the venue most likely to ensure visibility, relevance, and prestige for the publication, as well as reach the appropriate target audience. The venues we recommend are JAMA Network Open, PLOS One, and the Journal of the American Medical Informatics Association. Each of these three venues appeared in the top five recommendations for each model run, and further analysis showed a majority of their articles were predicted to be in the *High* category for number of citations. Our findings indicate that articles in these venues exhibit greater visibility, relevance, and prestige than those in the other venues in our dataset. When working with stakeholders, it is important to emphasize that research visibility provided by any venue is tied to results over time.

Next Steps

Based on our final conclusions, our next step would be to test the algorithms with more data and push for articles published between 2015 and 2022. More data would include new articles from any venue, both within and outside our dataset. This would allow us to better understand the venues we currently have and unknown ones that might offer critical visibility. More data gives us better insights, resulting in more accurate data-based decisions. We also strive to strengthen our algorithms to decrease any substantial temporal bias by training on articles published between 2015 and 2022. This ensures our data is within the same date range, includes techniques relevant to today's research, and removes extreme outliers in citation count accumulated over time. To accomplish this, we could work with external sources that provide article and venue information in a dataset, thus avoiding having to manually pull articles and instead have a clean database to retrieve them from. This would give our research and guidance a stronger foundation while also being more dynamic.

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